SHARP BOUNDS, STABILITY, AND ARITHMETIC FLUCTUATIONS FOR INDUCED PENTAGONS

Abstract. We prove that every graph with $m > 0$ edges contains at most

$$\frac{3}{2\sqrt{2}}\left(\frac{m}{5}\right)^{5/2}$$

induced pentagons. Equality holds precisely for balanced blow-ups of the Clebsch graph, together with isolated vertices. The bound and equality classification extend to arbitrary nonnegative vertex weights and to all graphons, including fractional graphons. Moreover, a deficit of ε in the normalized ratio can be repaired by $O(\sqrt{\varepsilon}\, m)$ edge edits on the original vertex set, producing an exactly balanced Clebsch blow-up and isolates. The exponent $1/2$ is optimal. We further separate the linear cost of adjacency errors from the quadratic cost of class imbalance. This yields an arithmetic description of the finite-edge deficit and its explicit second-order term, with a uniform power-saving remainder. For each fixed integer j, the exact extremal count at $m = 40k^2 + j$ is eventually an integer coefficient polynomial in k, computable by a finite enumeration; we do not quantify its threshold of validity. The proof combines global variational conditions, an active-density bound, and an exact rational certificate on the 1,044 graphs of order seven. A positive-slope identity permits a single rational reference density. An exact conditional span excludes triangles at equality, after which degree and codegree identities determine the Clebsch structure. All finite certificates admit independent verification using integer arithmetic.

1. Introduction and results

The edge version of inducibility asks how many induced copies of a fixed graph can occur in a graph with a prescribed number of edges. Recent work of Wang, Zhao, and Lu [13] and of Chao, Cohen Antonir, Li, and Yu [5] studies this problem through entropy inequalities and local directed graphs. For the pentagon, both works conjecture that balanced blow-ups of C_5 give the asymptotically optimal count $(m/5)^{5/2}$; see [13, Conjecture 8] and [5, Conjecture 1.7]. We determine the sharp constant, which is larger by a factor of $3/(2\sqrt{2})$, and prove the corresponding rigidity and stability statements. We then resolve the second-order arithmetic oscillation and

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obtain an eventual exact polynomial at each fixed offset from the equality sequence.

The extremal construction itself is known. Davey, Hurley, de Joannis de Verclos, Kang, and Volec [6, Lemmas 1.4, 7.1, and 7.2] record the Clebsch pentagon count and its balanced blow-ups while studying triangle-free pentagon counts normalized by $|G|\Delta(G)^4$. Our upper bound concerns arbitrary graphs with a prescribed edge count. We include the elementary construction count below to fix the normalization.

With the number of vertices prescribed, the induced-pentagon problem has a different extremal structure. Balogh, Hu, Lidický, and Pfender [1] established the asymptotic result using iterated C_5 blow-ups; Lidický, Mattes, and Pfender [7] subsequently determined the extremal count for every order. Those results provide context for the induced-count problem. Here the active vertex count is part of the optimization and may be accompanied by an arbitrary number of isolated vertices.

Let H be a finite simple graph with nonnegative vertex weights $x = (x_v)_{v\in V(H)}$. Define

$$E(H,x) = \sum_{uv\in E(H)} x_u x_v, \qquad C(H,x) = \sum_{\substack{U\subseteq V(H)\\ H[U]\cong C_5}} \prod_{v\in U} x_v.$$

Thus all pentagons counted by C are induced, including when H contains triangles. Set

$$c_* = \frac{3}{50\sqrt{10}} = \frac{192}{40^{5/2}}, \qquad R_* = \frac{3}{2\sqrt{2}} = 5^{5/2}c_*.$$

The Clebsch graph is the Cayley graph on $\mathbb{F}_2^4$ generated by

$$\{e_1, e_2, e_3, e_4, e_1+e_2+e_3+e_4\}.$$

It has degree five and forty edges. The only nonempty linear dependence among these five generators is their full sum. Every closed walk of length five therefore uses all five generators once and is a simple pentagon. There are $16\cdot 5!/10 = 192$ pentagons, and there are no triangles. Every pentagon is induced.

Theorem 1.1 (Sharp weighted bound and equality)**.** *Every finite simple graph with nonnegative vertex weights satisfies*

$$C(H,x) \le c_* E(H,x)^{5/2}. \tag{1}$$

If $E(H,x) > 0$, equality holds if and only if, after zero-weight vertices are discarded, the graph consists of a Clebsch blow-up and isolated vertices, with the sixteen Clebsch classes having equal total weight.

Write $I_5(G)$ for the number of induced pentagons in an ordinary graph. A graph with $m > 0$ edges satisfies

$$I_5(G) \le R_* \left(\frac{m}{5}\right)^{5/2}.$$

Equality holds exactly for a Clebsch blow-up with the same positive integer number k of vertices in every class, together with isolates. Such a graph has $40k^2$ edges and $192k^5$ induced pentagons.

Corollary 1.2 (Asymptotic edge inducibility)**.** *If $\rho(C_5, m)$ is the largest number of induced pentagons in a graph with exactly m edges, then*

$$\rho(C_5, m) = c_* m^{5/2} + O(m^2).$$

Proof. The upper bound is Theorem 1.1. For the lower bound, take a balanced Clebsch blow-up with $k = \lfloor \sqrt{m/40} \rfloor$ and add a disjoint matching with $m - 40k^2$ edges. It has exactly m edges and $192k^5$ induced pentagons. Since $m - 40k^2 = O(\sqrt{m})$, the asserted error follows. □

In the normalization of [5], our result reads

$$\mathrm{eind}(C_5) = \limsup_{m\to\infty} \frac{10\rho(C_5, m)}{(2m)^{5/2}} = \frac{3}{40\sqrt{5}}.$$

The upper bound in [5, Theorem 1.8] is $1/\sqrt{250}$ in this normalization. Thus the sharp bound also settles the pentagon case identified as open in that paper.

For a graphon $W : [0,1]^2 \to [0,1]$, use the normalization

$$E(W) = \frac{1}{2} t(K_2, W), \qquad C(W) = \frac{1}{10} t_{\mathrm{ind}}(C_5, W),$$
$$R(W) = 5^{5/2} \frac{C(W)}{E(W)^{5/2}}.$$

Let $\mathcal{E}_{\mathrm{Cl}}$ be the family obtained by partitioning an active set of arbitrary positive measure into sixteen equal classes, placing Clebsch adjacency between them, and making all remaining points isolated.

Theorem 1.3 (Graphon bound and equality)**.** *Every positive-density graphon satisfies $R(W) \le R_*$. Equality holds exactly for $\mathcal{E}_{\mathrm{Cl}}$, up to a measurable partition and null sets. In particular every equality graphon is $\{0,1\}$-valued.*

The stability statement is on the edge scale. If $I_5(G) \ge (R_* - \varepsilon)(m/5)^{5/2}$, then $O(\sqrt{\varepsilon}\, m)$ edge additions and deletions suffice to obtain an exactly balanced Clebsch blow-up and isolates on the original vertex set. Theorem 8.1 states this precisely. There is no bound on the original number of vertices and no additional rounding error. The exponent $1/2$ cannot be improved.

Our stronger structural conclusion distinguishes adjacency errors from class imbalance. A graph with normalized deficit ε can be edited into a possibly unbalanced Clebsch blow-up with integer class sizes n_i, total active size N, and

$$|E(G) \triangle E(H)| + \sum_{i=1}^{16} (n_i - N/16)^2 = O(\varepsilon m).$$

There is no additive rounding loss. We also prove a two-sided comparison between the pentagon deficit and this structural quantity; see Section 9.

The integer edge count produces a nonconstant second-order term. Writing

$$\alpha_m = \left\{ \sqrt{\frac{32m}{5}} \right\}, \qquad f(\alpha) = \frac{(1-\alpha)\min\{3\alpha, 1\}}{80},$$

Theorem 12.2 gives the uniform expansion

$$\rho(C_5, m) = c_* m^{5/2} - m^2 f(\alpha_m) + O(m^{15/8}). \tag{2}$$

In particular,

$$\limsup_{m\to\infty} \frac{c_* m^{5/2} - \rho(C_5, m)}{m^2} = \frac{1}{120}.$$

The phase function comes from two different constructions: a partial clone of a balanced template, and an integer rounding of a perturbation in the negative eigenspace of the Clebsch matrix. The lower bound reduces every extremal graph to the same scalar profile problem. Theorem 10.1 additionally determines the order of the deficit uniformly at phases approaching zero, where the absolute remainder in (2) need not be smaller than the displayed second-order term. At every fixed integer offset $m = 40k^2 + j$, Theorem 13.1 gives the finer expansion $c_* m^{5/2} - \rho(C_5, m) = \Lambda(j)k^3 + O_j(k^2)$, where $\Lambda(j)$ is an explicit finite lattice optimization. Theorem 14.1 then gives an eventually exact integer coefficient polynomial for every fixed offset, together with a finite procedure determining all eventual extremal constructions. For example, Corollary 13.2 proves, for all sufficiently large k,

$$\rho(C_5, 40k^2 + 1) = 192k^5, \qquad \rho(C_5, 40k^2 - 1) = 192k^5 - 24k^3.$$

Our certificate uses the flag algebra framework of Razborov [12]. Strong and perfect blow-up stability from flag algebra calculations has a general fixed-order framework due to Pikhurko, Sliačan, and Tyros [11]. Bodnár and Pikhurko [4] combine exact flag algebra bounds with stability and integer optimization of blow-up part sizes in several inducibility problems. Their (κ, ℓ)-edge-inducibility objective counts κ-vertex subsets with exactly ℓ edges in a graph of fixed order; it is different from prescribing the total edge count of the host graph, as done here. The systematic study of six-vertex inducibility by Bodnár, Gao, León, Liu, Pikhurko, and Sun [3] provides further instances of sharp flag algebra bounds and perfect stability. Thus the use of exact certificates and the reduction of an extremal problem to integer part-size optimization are established techniques. The estimates here are measured by m, allow the active mass to vary, and control integer class variance separately from adjacency errors. These features are retained in the arithmetic analysis, which also allows bounded exceptional edge modifications at each fixed offset.

Automated numerical optimization and conversion to exact proofs are available in Bodnár's FlagAlgebraToolbox [2]. The computations for this paper use the separately supplied generators and exact verifiers; FlagAlgebraToolbox is cited as a software and methodological precedent, not as a dependency of these computations.

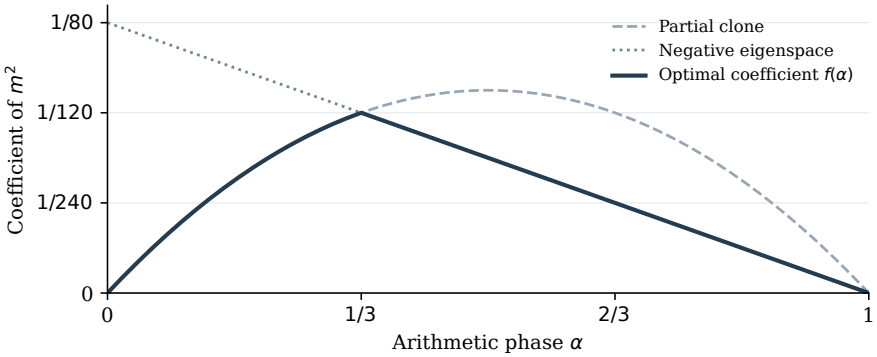


FIGURE 1. The second-order coefficient and the two constructions that attain it. Their transition is at $\alpha = 1/3$, where $f(\alpha) = 1/120$.

The proof first constructs a global graphon maximum. Global optimality supplies vertex, edge, and new-profile variations; these are stronger than weight optimality on one fixed template. A density bound allows every relevant global maximum, including a fractional one, to be padded to $p_* = 5/144$. An exact seven-vertex identity then proves the upper bound at that density. A second exact span excludes triangles at equality. Restrictions of the same positive-square vectors then give the degree and codegree structure and identify the Clebsch graph. Finally, peeling, compactness, local L^1 stability, and whole-vertex rounding give the quantitative edit bound. Keeping the linear adjacency defect and the quadratic mass defect separate then permits the finite-edge arithmetic analysis.

2. Global extremizers and their variations

It is useful to allow a finite vertex measure μ, of total mass $s = \mu(\Omega)$, and a symmetric measurable kernel $W : \Omega^2 \to [0, 1]$. All densities in this section are integrated against μ. Thus

$$E = \frac{1}{2}\int W(x,y)\,d\mu(x)\,d\mu(y),$$
$$C = \frac{1}{10}\int \prod_{ij\in E(C_5)} W(x_i,x_j) \prod_{\substack{i<j\\ ij\notin E(C_5)}} (1 - W(x_i,x_j))\,d\mu^5.$$

Multiplying μ by a constant does not change $C/E^{5/2}$. Write $d(x) = \int W(x,y)\,d\mu(y)$ and let $L(x)$ be the first derivative of C with respect to vertex mass at x. Equivalently, $L(x)$ is one half of the induced rooted pentagon integral on four further vertices. A rooted pentagon is always induced throughout the paper.

2.1. **Finite templates and a compact maximization problem.**

Lemma 2.1 (Finite-template attainment)**.** *For a fixed finite graph containing an induced pentagon, the supremum of $C(H,x)/E(H,x)^{5/2}$ over nonnegative weights with $E(H,x)>0$ is attained on a positive-weight induced subgraph.*

Proof. Normalize $E=1$. The square of a pentagon monomial is the product of its five cyclic edge weights, and is at most 5^{-5} by AM–GM. There are finitely many such monomials, so the supremum is finite. Take a maximizing sequence and pass to a subsequence on which every vertex weight converges in $[0,\infty]$.

If a pentagon contains a weight $M\to\infty$, its two cyclic neighbors have weights at most $1/M$. Its remaining two vertices are adjacent, so their product is at most one. The pentagon monomial is consequently at most $1/M$ and tends to zero. A monomial with only finite limiting coordinates, one of them zero, also tends to zero. The finite positive limiting coordinates therefore retain the entire limiting pentagon mass. Their edge mass is at most one and is positive. If it were strictly less than one, their ratio would exceed the original supremum. Hence it equals one, proving attainment. □

Lemma 2.2 (A global graphon maximum)**.** *The supremum of R over all finite graphs, over all finite nonnegative weighted graphs, and over all positive-density graphons is the same finite number. It is attained by a graphon of edge density* $1/80$*.*

Proof. The finite ordinary and weighted suprema agree. Indeed, rational weights are realized by integer independent blow-ups, and an induced C_5 cannot use two vertices of one blow-up class: those vertices would be false twins, whereas C_5 has no false twins. Approximation by rational weights gives the claim.

At a positive finite-template maximum, remove isolates and normalize the weights to sum one. Put

$$p=2E,\qquad \theta=10C,\qquad \kappa=\frac{5C}{2E}=\frac{\theta}{2p}.$$

The first variation is $C_i=\kappa d_i$. The elementary universal link inequality

$$C_i\le\frac{1}{4}Ed_i^2 \tag{3}$$

is proved, including its fractional version, in Lemma 3.1. Since all active degrees are positive, it implies $d_i\ge 4\theta/p^2$. Averaging yields

$$p\ge\frac{16R^2}{1000},\qquad R=\sqrt{1000}\,\frac{\theta}{p^{5/2}}. \tag{4}$$

In particular the common finite supremum M is at most $\sqrt{1000}/4$, and it is at least R_* by the Clebsch construction.

Every positive-density graphon has ratio at most M. To see this, sample independent latent points and then sample edges independently with the specified probabilities. For any fixed induced pattern, the normalized injective count converges in probability to its graphon density. The variance is $O(1/N)$:

pairs of injections on disjoint vertex sets are independent, and overlapping pairs form an $O(1/N)$ fraction. Apply this simultaneously to the edge and induced pentagon counts. The edge density has a positive limit, so the ratios converge as well.

Conversely, every finite graph with ratio at least R_* can first be optimized on its finite template. By (4), the resulting active probability weighting has edge density at least $9/500 > 1/80$. Rescale its active measure and add isolates to obtain edge density $1/80$, without changing its ratio. Thus the supremum over graphons of this fixed density is also M.

We use the compactness of graphons modulo cut distance [9, Theorem 5.1] and the continuity of fixed homomorphism densities [8]. The induced pentagon density is continuous as well: expand its five nonedge factors into a finite alternating sum of ordinary homomorphism densities. The class of graphons of edge density $1/80$ is closed and compact, and the denominator of R is fixed there. It therefore contains a maximizer. The sampling argument shows that this is a global maximum over all positive edge densities, not only over the fixed-density class. □

This globality is essential. Edge variations and insertion of arbitrary new vertex profiles cannot be inferred merely from optimality of the weights on a fixed finite template.

2.2. **Vertex and edge variations.** Let $A(x,y)$ be the induced pentagon completion kernel after one cyclic edge factor is deleted, and let $B(x,y)$ be the completion kernel after one chord nonedge factor is deleted. For explicit conventions, suppressing the measure symbols,

$$\begin{aligned} A(x,y) &= \int W(x,a)W(a,b)W(b,c)W(c,y)\,N_A\,da\,db\,dc,\\ N_A &= (1-W(x,b))(1-W(x,c))(1-W(y,a))\\ &\quad\cdot(1-W(y,b))(1-W(a,c)),\\ B(x,y) &= \int W(x,a)W(a,y)W(y,b)W(b,c)W(c,x)\,N_B\,da\,db\,dc,\\ N_B &= (1-W(x,b))(1-W(a,b))(1-W(a,c))(1-W(y,c)). \end{aligned}$$

Both kernels are symmetric. Put $q = A - B$.

Lemma 2.3 (Necessary conditions at a global maximum)**.** *At a positive global graphon maximum of $C/E^{5/2}$, define*

$$\kappa = \frac{5C}{2E}, \quad C'' = WA + (1-W)B, \quad M = C'' - \kappa W, \quad G = \frac{3\kappa}{2E}d\otimes d - M.$$

Then

$$L(x) = \kappa d(x) \qquad \text{almost everywhere}, \tag{5}$$

$$G \succeq 0 \qquad \text{as an integral operator on } L^2(\mu), \tag{6}$$

$$W(q-\kappa) \geq 0, \quad (1-W)(q-\kappa) \leq 0 \qquad \text{almost everywhere}. \tag{7}$$

Consequently

$$M \geq B \geq 0, \qquad \int W(x,y)M(x,y)\,d\mu(y) = \kappa d(x) \quad \text{almost everywhere.} \tag{8}$$

Proof. Replace $d\mu$ by $(1 + sh)\,d\mu$, with h bounded and s of either sign sufficiently small. Normalizing the total measure afterward does not change the ratio. The first derivative gives (5); the second gives (6), first for bounded h and then for all $h \in L^2(\mu)$ by density. The kernel of the second vertex derivative is $WA + (1-W)B$: the two selected cycle vertices are adjacent in ten ordered positions and nonadjacent in ten ordered positions, canceling the factor $1/10$ in C.

For a bounded symmetric edge perturbation H,

$$DC(W)[H] = \frac{1}{2}\int q(x,y)H(x,y)\,d\mu^2, \qquad DE(W)[H] = \frac{1}{2}\int H(x,y)\,d\mu^2.$$

Use the feasible one-sided directions $H = -Wf$ and $H = (1-W)f$, with $f \geq 0$. Global maximality yields (7). In particular $W(1-W)(q-\kappa) = 0$ almost everywhere.

The identity $M = B + W(q-\kappa)$ proves $M \geq B$. Furthermore

$$\int W(x,y)A(x,y)\,d\mu(y) = 2L(x).$$

Using $C'' = WA + (1-W)B$ and the preceding complementary slackness identity, we obtain

$$\begin{aligned}\int WM &= \int WA - \int W(1-W)q - \kappa\int W^2 \\ &= 2L - \kappa d = \kappa d.\end{aligned}$$

This proves (8), including on fractional parts of the graphon. □

On a probability space write

$$p = t(K_2, W), \qquad \theta = t_{\mathrm{ind}}(C_5, W).$$

Multiplying (5) by d, d^2, and $b(x) = \int W(x,y)d(y)\,dy$, respectively, gives

$$p\,t(L_j, W) = \theta\,t(B_j, W), \qquad j = 1,2,3. \tag{9}$$

Here L_1, L_2, L_3 are an induced pentagon with, respectively, a leaf, two leaves at the same root, and a pendant path of length two at that root. Only the five original cycle vertices are required to induce a pentagon; all other unmentioned pairs are unrestricted. The graphs B_1, B_2, B_3 are $P_3, K_{1,3}$, and P_4, with ordinary homomorphism density conventions.

2.3. **Insertion of a new vertex profile.** For a measurable function $f : \Omega \to [0,1]$, write $d(f) = \int f$ and

$$L(f) = \frac{1}{2}\int I_{P_4}(a,b,c,d)f(a)f(d)(1-f(b))(1-f(c))\,d\mu^4, \tag{10}$$

where

$$I_{P_4} = W(a,b)W(b,c)W(c,d)(1-W(a,c))(1-W(a,d))(1-W(b,d)).$$

Lemma 2.4 (Profile insertion)**.** *At a positive global maximum, $L(f) \le \kappa d(f)$ for every measurable $f : \Omega \to [0,1]$.*

Proof. Add a new independent class of mass ϵ, with adjacency profile f to the old space. Before normalizing its total measure, the new masses are

$$E_\epsilon = E + \epsilon d(f), \qquad C_\epsilon = C + \epsilon L(f) + O(\epsilon^2).$$

The linear term selects exactly one new cycle vertex. If f is fractional, cycles using two new vertices can occur, but their contribution is still $O(\epsilon^2)$. Differentiating the ratio in this feasible direction proves the claim. □

In particular, put $g = W(u,\cdot)$, $h = W(v,\cdot)$, and $f = g(1-h)$. Combining the lemma with (5) gives

$$J_\theta(u,v) := L(g) - L(f) - \frac{\theta}{2p}(d(g) - d(f)) \ge 0 \tag{11}$$

on a probability space. Its removed degree $d(g) - d(f) = \int gh$ is nonnegative. Multiplying (11) by the nonnegative factor $W(u,v)W(u,z)W(v,z)$ and integrating gives the one profile condition used in the certificate.

Lemma 2.5 (Padding)**.** *Restrict a probability graphon to its positive-degree set and normalize that restriction to have total measure one. If its edge density is $p_a > p_*$, give this normalized active graphon total measure $q = \sqrt{p_*/p_a}$ and adjoin isolate mass $1-q$. The resulting graphon has the original ratio R and edge density p_*. If the original graphon is a global ratio maximum, so is the padded one, and all the preceding necessary conditions apply.*

Proof. Edge mass and pentagon mass scale by the second and fifth powers of the active measure, respectively. Their normalized ratio is unchanged. Globality follows from equality of the ratio with the global supremum, rather than from any restricted variational argument on the original template. □

3. A lower bound on the active density

We work temporarily on a finite measure space (Ω, μ) and write $S = \mu(\Omega)$. The kernel W always takes values in $[0,1]$; rescaling the vertex measure changes neither the ratio $C/E^{5/2}$ nor its global maximality. Throughout this section,

$$d(x) = \int_\Omega W(x,y)\,d\mu(y), \qquad b(x) = \int_\Omega W(x,y)d(y)\,d\mu(y),$$

$$S_2 = \int_\Omega d(x)^2\,d\mu(x).$$

All integrals below are with respect to the indicated powers of μ. The vertex derivative of C is denoted by L_x, and $\kappa = 5C/(2E)$.

Lemma 3.1 (Universal link bounds)**.** *For every $[0,1]$-valued graphon on a finite measure space, set*

$$\begin{aligned} g_x(u) &= W(x,u), \\ s_x &= \int_{\Omega^2} g_x(u)(1-g_x(v))W(u,v)\,d\mu(u)\,d\mu(v), \\ E_{\mathrm{out}}(x) &= \frac{1}{2}\int_{\Omega^2}(1-g_x(u))(1-g_x(v))W(u,v)\,d\mu(u)\,d\mu(v). \end{aligned}$$

Then

$$L_x \le \frac{d(x)^2}{4} E_{\mathrm{out}}(x) \le \frac{E\,d(x)^2}{4}, \tag{12}$$

and

$$L_x \le \frac{s_x^2}{2}, \qquad E_{\mathrm{out}}(x) = E - \frac{b(x)+s_x}{2}. \tag{13}$$

Proof. Fix x and abbreviate g_x to g. If

$$\Phi(a,b,c,e) = W(a,b)W(b,c)W(c,e)(1-W(a,c))(1-W(a,e))(1-W(b,e)),$$

then the exact rooted link formula is

$$L_x = \frac{1}{2}\int_{\Omega^4} \Phi(a,b,c,e)g(a)g(e)(1-g(b))(1-g(c)).$$

The factor $1/2$ identifies the two orientations of the rooted pentagon. For a fixed opposite edge bc, the two endpoint choices have masses

$$\begin{aligned} U_{bc} &= \int_{\Omega} g(a)W(a,b)(1-W(a,c))\,d\mu(a), \\ V_{bc} &= \int_{\Omega} g(a)W(a,c)(1-W(a,b))\,d\mu(a). \end{aligned}$$

The elementary inequality $u(1-v)+v(1-u) \le 1$ for $u,v \in [0,1]$ gives $U_{bc}+V_{bc} \le d(x)$. Dropping the nonedge factor between the two endpoints and using $U_{bc}V_{bc} \le d(x)^2/4$ proves the first inequality in (12). The second follows from $E_{\mathrm{out}}(x) \le E$.

For the other bound, retain in the link integrand only

$$g(a)(1-g(b))W(a,b)\; g(e)(1-g(c))W(e,c).$$

All discarded factors lie in $[0,1]$. The two remaining pairs integrate independently, giving $L_x \le s_x^2/2$. Finally, with

$$t_x = \frac{1}{2}\int_{\Omega^2} g(u)g(v)W(u,v),$$

we have $b(x) = 2t_x + s_x$ and $E_{\mathrm{out}}(x) = E - b(x) + t_x$. These are exactly the last identity in (13). □

The next estimate uses global maximality. We recall the notation from the variation formulas: A and B are the completions of an induced pentagon after deleting, respectively, the root edge and the root nonedge. Put

$$M = C'' - \kappa W = B + W(A - B - \kappa).$$

At a global maximizer the first-order, edge, and second-order conditions give

$$\begin{gathered} L_x = \kappa d(x), \qquad M(x,y) \geq B(x,y) \geq 0, \\ \int_\Omega W(x,y)M(x,y)\,d\mu(y) = \kappa d(x), \\ G = \frac{3\kappa}{2E} d \otimes d - M \succeq 0. \end{gathered} \tag{14}$$

In particular, after zero-degree points are discarded, Lemma 3.1 implies

$$d(x) \geq \frac{4\kappa}{E} = \frac{10C}{E^2} \quad \text{almost everywhere.} \tag{15}$$

Lemma 3.2 (A second-order moment inequality)**.** *At a global maximizer with $C > 0$, on its active finite measure space,*

$$S_2 \geq \frac{ES}{3} + \frac{20C}{E}. \tag{16}$$

Proof. Write

$$\gamma(x,y) = \int_\Omega W(x,z)W(y,z)\,d\mu(z)$$

for codegree, and let

$$Q(x,y) = \int_{\Omega^2} W(x,a)W(a,b)W(b,y)(1-W(x,b))(1-W(a,y))\,d\mu(a)\,d\mu(b).$$

Thus Q is the completion of an induced three-edge path with its endpoint nonedge factor omitted. In $B(x,y)$, delete the two nonedge factors joining the common neighbor to the internal vertices of this path. This gives

$$B(x,y) \leq \gamma(x,y)Q(x,y).$$

Define

$$D = \int_{\Omega^2} (1-W)Q = t_{\mathrm{ind}}(P_4, W), \qquad L_0 = \int_{\Omega^2} (1-W)B\gamma.$$

Since $\int (1-W)B = 10C$, Cauchy–Schwarz yields

$$100C^2 \leq DL_0. \tag{17}$$

Here B/γ may be assigned the value zero where $\gamma = 0$; the bound $B \leq \gamma Q$ makes this convention harmless.

We also have

$$D \leq E^2. \tag{18}$$

Indeed, fix the two outer edges on four independent points. Of the four possible cross edges, the probabilities that exactly one is present sum to at most one. Each choice, after integration, is the same induced-P_4 density D.

Consequently $4D \le (\int W)^2 = 4E^2$. This proof applies directly to fractional values of W.

Let T be the integral operator with kernel W. We now justify an optimized diagonal test of the positive operator G in (14). Fix a bounded real function z, and choose increasing finite measurable partitions whose conditional expectations of W, M, d, z converge in L^2, using the product partitions for the two kernels. Such partitions exist by approximating the kernels by finite linear combinations of measurable rectangles and the functions by simple functions, then successively refining all the sets that occur. Let P_n denote orthogonal projection onto the associated step functions, and set

$$T_n = P_n T P_n, \qquad M_n = P_n M P_n, \qquad d_n = P_n d.$$

These are compressions of the original kernels, not derivatives recomputed at an approximating graphon. In particular

$$G_n = \frac{3\kappa}{2E} d_n \otimes d_n - M_n \succeq 0.$$

Put $z_n = P_n z$ and let Z_n be multiplication by z_n on the finite-dimensional step space. The matrix $(T_n - Z_n)^2$ is positive semidefinite, so

$$\operatorname{tr}\bigl(M_n (T_n - Z_n)^2\bigr) \le \frac{3\kappa}{2E} \|(T_n - Z_n) d_n\|_2^2.$$

In the orthonormal basis $\mathbf{1}_U/\sqrt{\mu(U)}$ of partition cells, each diagonal entry of M_n is nonnegative because $M \ge 0$. Therefore $\operatorname{tr}(M_n Z_n^2) \ge 0$, and discarding this term gives

$$\operatorname{tr}(M_n T_n^2) \le \frac{3\kappa}{2E} \|T_n d_n - Z_n d_n\|_2^2 + 2 \operatorname{tr}(M_n T_n Z_n). \tag{19}$$

We explain all three limits. Conditional expectations of the bounded kernels converge in Hilbert–Schmidt norm, so $\operatorname{tr}(M_n T_n^2) \to \operatorname{tr}(M T^2)$. Also $T_n d_n - Z_n d_n \to b - z d$ in L^2: the kernels, d_n, and z_n are uniformly bounded, and the conditional expectations converge in L^2. Finally, if the same symbols denote the step kernels, the mixed trace is

$$\operatorname{tr}(M_n T_n Z_n) = \int_{\Omega^2} z_n(x) M_n(x,y) W_n(x,y)\, d\mu(x)\, d\mu(y).$$

The integrands converge in L^1, and (14) identifies the limit as $\kappa \int z d$. No infinite-dimensional trace involving the arbitrary multiplication operator Z^2 is required; its nonnegative contribution was discarded before taking a limit.

The kernel of T^2 is $\gamma \ge 0$. Since $M \ge B$,

$$\operatorname{tr}(M T^2) = \int_{\Omega^2} M\gamma \ge \int_{\Omega^2} (1 - W) B \gamma = L_0.$$

Passing to the limit in (19) gives

$$L_0 \le \kappa \int_\Omega \left[\frac{3}{2E} (b - zd)^2 + 2zd \right] d\mu. \tag{20}$$

Choose

$$z(x) = \frac{b(x) - 2E/3}{d(x)}.$$

This is bounded: $b/d \le S$, and the positive lower bound (15) controls $1/d$ on the active space. The choice minimizes the scalar quadratic in (20), yielding

$$L_0 \le \kappa \left(2 \int_\Omega b \, d\mu - \frac{2ES}{3} \right) = 2\kappa \left(S_2 - \frac{ES}{3} \right).$$

Combine this with (17) and (18), and use $\kappa = 5C/(2E)$. The result is (16). □

Proposition 3.3 (Active density)**.** *Let W be a global maximizer of $C/E^{5/2}$ among graphons of positive edge density. Suppose*

$$5^{5/2} \frac{C}{E^{5/2}} \ge \frac{3}{2\sqrt{2}}.$$

After its zero-degree set is deleted and the remaining measure is normalized to one, its edge density satisfies

$$p_{\mathrm{act}} \ge \frac{80000}{(491\sqrt{10} - 75\sqrt{2})^2} > \frac{5}{144}. \tag{21}$$

Proof. Delete zero-degree points, which changes neither E nor C, and rescale the measure so that $E = 1$. Write $c = C$ in this normalization, so $c \ge c_* = 3/(50\sqrt{10})$ and $\kappa = 5c/2$. By (15),

$$d(x) \ge \delta := 10c, \qquad S \le \frac{1}{5c}. \tag{22}$$

The second inequality uses $\int d = 2$.

Put

$$T_0 = \int_\Omega d(x) b(x) \, d\mu(x), \qquad V = \int_\Omega d(x) s_x \, d\mu(x).$$

The inequality $d(x)d(y) \ge \delta(d(x) + d(y)) - \delta^2$, integrated against $W(x, y)$, and Lemma 3.2 give

$$T_0 \ge 2\delta S_2 - 2\delta^2 \ge \frac{20}{3} cS + 200c^2. \tag{23}$$

Here δ is the fixed bound $10c$, so no substitution is made in a nonmonotone function of the actual minimum degree. By $L_x = \kappa d(x)$ and (13), $s_x \ge \sqrt{2\kappa d(x)}$. Jensen's inequality therefore gives

$$V \ge \sqrt{2\kappa} \int_\Omega d(x)^{3/2} \, d\mu(x) \ge 4\sqrt{\frac{\kappa}{S}}. \tag{24}$$

Divide the first inequality in (12) by $d(x) > 0$, use $L_x = \kappa d(x)$, and integrate. With $E_{\mathrm{out}}(x) = 1 - (b(x) + s_x)/2$ this reads

$$8\kappa S + T_0 + V \le 4.$$

Equations (23) and (24) now imply

$$\frac{40}{3} cS + 100c^2 + 2\sqrt{\frac{\kappa}{S}} \le 2.$$

The mass bound in (22) gives $2\sqrt{\kappa/S} \geq 5\sqrt{2}\,c$, and hence

$$S \leq \frac{3}{20c} - \frac{15}{2}c - \frac{3\sqrt{2}}{8}.$$

The right-hand side is strictly decreasing for $c > 0$. Substituting $c_* = 3/(50\sqrt{10})$ yields

$$S \leq \frac{491\sqrt{10} - 75\sqrt{2}}{200}.$$

Since $11\sqrt{10} < 75\sqrt{2}$, this upper bound is strictly less than $480\sqrt{10}/200 = \sqrt{288/5}$. On normalizing the active measure to one, its edge density is $p_{\mathrm{act}} = 2/S^2$. This proves both inequalities in (21). □

In particular, a global maximizer can be padded by isolated mass to the rational density $p_* = 5/144$. The same conclusion applies to any equality graphon once the sharp bound has been established: such a graphon is itself a global maximizer. Thus the padding needed for the equality analysis does not require a finite-support or a $\{0,1\}$-valued assumption.

4. A single exact certificate

We work on a probability space and write

$$p = t(K_2, W), \qquad \theta = t_{\mathrm{ind}}(C_5, W), \qquad p_* = \frac{5}{144}, \qquad \tau = \frac{5}{663552}.$$

Balanced Clebsch on an active set of measure $1/3$, with isolates outside it, has $(p, \theta) = (p_*, \tau)$. The certificate in this section applies to arbitrary graphons of density p_*; its variational interpretation is used at a global maximum.

4.1. **The additional variational columns.** Retain the partially induced patterns L_j and the graphs B_j from (9), and abbreviate

$$\Phi_j(s) = p_* t(L_j, W) - s\, t(B_j, W), \qquad j = 1, 2, 3.$$

We retain four additional columns. We describe them explicitly to specify their signs and their dependence on the target density. Two have zero multipliers in the supplied certificate.

First let $z(x) = 2L(x)$ be the rooted labeled induced-pentagon density, and put

$$T(x) = \int W(x,y)W(x,w)W(y,w)\,dy\,dw,$$

$$L_\triangle = \int z(x)T(x)\,dx, \qquad B_\triangle = \int d(x)T(x)\,dx.$$

Equivalently, $L_\triangle$ is the partially induced pattern obtained by attaching a triangle at one vertex of an induced pentagon, and $B_\triangle$ is the ordinary density of a triangle with a leaf at its root. Define

$$\Phi_\triangle(s) = p_* L_\triangle - s B_\triangle.$$

Multiplying the first-order identity by $T(x)$ gives $\Phi_\triangle(\theta) = 0$, and $B_\triangle \geq 0$.

For the next two columns, fix ordered roots u, v, write $g = W(u,\cdot)$ and $h = W(v,\cdot)$, and set

$$\psi_0(u,v) = \int g(x)h(x)g(y)h(y)(1 - W(x,y))\,dx\,dy,$$
$$\psi_1(u,v) = \int g(x)h(x)g(y)h(y)W(x,y)\,dx\,dy.$$

Conditional on the root edge being present, these are the probabilities of a diamond with its two degree-three vertices rooted, and of a complete four-vertex graph. Using the completion kernels A, B from Section 2, define

$$J_i(s) = \int W(u,v)\psi_i(u,v)\left(A(u,v) - B(u,v) - \frac{s}{2p_*}\right)du\,dv, \qquad i = 0, 1,$$

and $R_i = \int W(u,v)\psi_i(u,v)\,du\,dv \geq 0$. The existing-edge condition (7) gives $J_i(\theta) \geq 0$ at a global maximum.

Finally, let $J_{\mathrm{del}}(s)$ be the integral of the expression in (11), with θ replaced by s, against $W(u,v)W(u,w)W(v,w)\,du\,dv\,dw$. Thus $J_{\mathrm{del}}(\theta) \geq 0$. Its removed-degree coefficient is

$$D_{\mathrm{del}} = \int W(u,v)c(u,v)^2\,du\,dv \geq 0, \qquad c(u,v) = \int W(u,x)W(v,x)\,dx.$$

In particular $D_{\mathrm{del}} = R_0 + R_1$. At fixed edge density,

$$\begin{aligned} &\Phi_j(\tau) = (\theta - \tau)t(B_j, W), \quad \Phi_{\triangle}(\tau) = (\theta - \tau)B_{\triangle}, \\ &J_i(\tau) = J_i(\theta) + \frac{\theta - \tau}{2p_*}R_i, \quad J_{\mathrm{del}}(\tau) = J_{\mathrm{del}}(\theta) + \frac{\theta - \tau}{2p_*}D_{\mathrm{del}}. \end{aligned} \tag{25}$$

The first line uses the vertex first-order conditions. All added coefficients of $\theta - \tau$ in the second line are nonnegative.

4.2. **Seven-vertex coefficients and flag squares.** There are 1044 unlabeled simple graphs on seven vertices. Their canonical adjacency masks, edge lists and orbit sizes are supplied with the certificate. Write $\pi_H(W)$ for the probability that seven independent graphon samples induce H. For a pattern on $r \leq 7$ labeled vertices, specify its required edges and nonedges; all other pairs are unrestricted. Its coefficient on H is the number of injections satisfying these requirements divided by $(7)_r$. Averaging this coefficient against $\pi_H(W)$ gives its graphon density exactly. In particular the induced-pentagon coefficient is $I_5(H)/252$.

The five rooted flag spaces are as follows.

Root type	Roots	Flag order	Flags	Denominator
One root	1	4	20	140
Three independent roots	3	5	72	1260
Edge and isolated root	3	5	72	1260
Path $01, 12$	3	5	72	1260
Triangle	3	5	72	1260

Root labels are fixed individually. Isomorphisms may permute only the extension vertices. For each host, sample an ordered root embedding and two disjoint extension sets. Cross-edges between the two extensions are unrestricted. The sampling denominators are $7\binom{6}{3} = 140$ and $(7)_3\binom{4}{2} = 1260$.

For a flag vector v, let $q_F(x)$ be its conditional flag probabilities at a root tuple and let $P_\sigma(x)$ be the probability of the root type. The associated square is

$$Q_v(W) = \int P_\sigma(x) \left(\sum_F v_F q_F(x) \right)^2 dx \geq 0. \tag{26}$$

The root-type factor occurs once. This convention is essential for fractional graphons. An individual finite host matrix need not be positive semidefinite; its average over $\pi_H(W)$ is the matrix represented by (26).

For clarity, the raw source counts use the injection denominator $(7)_r$ for a pattern on r labeled vertices. Thus the cycle and house completions in J_i, and $L_\triangle$, use 5040; R_i, $B_\triangle$, and D_{del} use 840. The link difference in J_{del} uses $10080 = 2 \cdot 7!$, including the $1/2$ in the rooted link (10). Internal symmetries permit cancellations in some of these fractions, but the data files retain the raw injection counts. Each expanded profile term is multilinear in the formal graph-edge variables, so Boolean-host averaging is valid also for fractional W.

Proposition 4.1 (Exact finite identity)**.** *For every graphon of density p_* there is the identity*

$$\begin{aligned} \tau - \theta - \sum_{j=1}^{3} \lambda_j \Phi_j(\tau) - \lambda_\triangle \Phi_\triangle(\tau) - \sum_{i=0}^{1} a_i J_i(\tau) - a_{\mathrm{del}} J_{\mathrm{del}}(\tau) \\ = \sum_{r=1}^{184} b_r Q_{v_r}(W) + \sum_H s_H \pi_H(W). \end{aligned} \tag{27}$$

Here $\lambda_\triangle, a_{\mathrm{del}}, b_r$ are strictly positive rational numbers, $a_0 = a_1 = 0$, and $s_H \geq 0$. Exactly 826 *host slacks are positive. The three unrestricted first-order multipliers satisfy the exact inequalities*

$$-\frac{1157}{10} < \lambda_1 < 0, \qquad \lambda_3 < 0, \qquad \lambda_2 + \lambda_3 > 310. \tag{28}$$

All rational coefficients and integer flag vectors are specified in `all_graph_exact_certificate.json`*.*

Proof. The supplied identity has one row for each of the 1044 hosts. Besides the displayed columns, its finite form includes the 34 unrestricted fixed-density identities $p_F(W)t(K_2, W) - p_* p_F(W)$, one for each unlabeled graph F on five vertices, where p_F is its induced-graph probability. Their sum is $t(K_2, W) - p_*$, so this basis is equivalent to that single identity together with any 33 of the five-vertex identities. In the finite coefficients the additional

edge is sampled disjointly from the vertices of F. These columns vanish after graphon averaging at density p_*.

The checker computes the entire host atlas, all five flag matrices, the partially induced patterns and all the additional columns from their edge requirements. It then computes every declared source column and verifies the 1044 rational identities and every required coefficient sign exactly. Clearing one common denominator reduces the final identity check to integer arithmetic. The rational comparisons in (28) are checked in the same way. Thus averaging the finite identities proves (27); the numerical search used to find the coefficients is not part of the verification. □

4.3. **A uniformly positive slope.** Put $x = t(P_3, W)$, $y = t(K_{1,3}, W)$ and $z = t(P_4, W)$. For every graphon,

$$z = \int W(u,v)d(u)d(v)\,du\,dv \le \frac{1}{2}\int W(u,v)\big(d(u)^2 + d(v)^2\big)\,du\,dv = y,$$

and Cauchy–Schwarz gives $x^2 \le py$. Consequently

$$\begin{aligned} 1 + \lambda_1 x + \lambda_2 y + \lambda_3 z &\ge 1 + \lambda_1 x + (\lambda_2 + \lambda_3)y \\ &\ge 1 + \lambda_1 x + \frac{\lambda_2 + \lambda_3}{p_*}x^2 \\ &\ge 1 - \frac{p_*\lambda_1^2}{4(\lambda_2 + \lambda_3)} > \frac{2232551}{3571200} > \frac{5}{8}. \end{aligned} \tag{29}$$

The penultimate comparison uses (28).

Corollary 4.2 (Bound at a global maximum)**.** *A positive global maximum of the induced-pentagon to edge ratio, normalized to edge density p_*, satisfies $\theta \le \tau$. If $\theta = \tau$, all* 184 *ordinary squares in (27) vanish.*

Proof. Insert (25) into (27). The resulting identity is

$$\begin{aligned} (\tau - \theta)\mathcal{B}(W) = &\sum_r b_r Q_{v_r}(W) + \sum_H s_H \pi_H(W) \\ &+ \sum_{i=0}^{1} a_i J_i(\theta) + a_{\mathrm{del}} J_{\mathrm{del}}(\theta), \end{aligned}$$

where

$$\mathcal{B}(W) = 1 + \sum_j \lambda_j t(B_j, W) + \lambda_\triangle B_\triangle + \frac{a_0 R_0 + a_1 R_1 + a_{\mathrm{del}} D_{\mathrm{del}}}{2p_*}.$$

The last two contributions to $\mathcal{B}$ are nonnegative, so (29) gives $\mathcal{B} > 5/8$. Every term on the right is nonnegative at a global maximum. This proves the bound and, at equality, the vanishing of each ordinary square, since all b_r are strictly positive. □

4.4. **Exact spans for equality.** We record three finite consequences of this same certificate. Each is an explicit rational linear combination, checked coordinate by coordinate; no numerical rank assertion is used.

Lemma 4.3 (Triangle domination)**.** *In the path type with root edges* $01, 12$ *and extension vertices* $3, 4$*, there is a vector* G *in the rational span of its* 53 *strictly positive square vectors such that*

$$G(F) = 0 \quad \text{if } F \text{ is triangle-free}, \qquad G(F) \geq 1 \quad \text{if } F \text{ contains a triangle.}$$

There are 24 *coordinates of the first kind and* 48 *of the second.*

Proof. The combination is specified in the supplied file

`all_graph_path3_triangle_domination_span.json`.

Its checker computes which of the 72 flags contain a triangle, checks that every source vector comes from a strictly positive square of the final certificate, and adds the rational combination in every coordinate. The result is zero on all 24 triangle-free coordinates and at least one on all other coordinates. In the supplied witness, G is itself a positively weighted source vector: it is the indicator of the 48 triangle-containing flags. □

Once triangle-freeness has been established, only triangle-free flag coordinates contribute to a conditional flag distribution. The following two span identities concern the restrictions of the same all-graph square vectors to those coordinates.

Lemma 4.4 (Degree span)**.** *In the triangle-free flags of the path type* $01, 12$*, the vector*

$$g(F) = 24\big(e_F(1,3) + e_F(1,4)\big) - 5 \tag{30}$$

lies in the rational span of the restricted strictly positive square vectors of that type.

Proof. The witness in `all_graph_tf_degree_codegree_spans.json` uses 22 nonzero coefficients from the 53 path-type sources. The independent checker identifies the 24 triangle-free flags from their edge sets and verifies (30) in all 24 coordinates by rational addition. All source squares have strictly positive rational weights. □

Lemma 4.5 (Codegree span)**.** *Consider the triangle-free flags of the type with the single root edge* 01 *and isolated root* 2*. Put*

$$h_i(F) = 24\big(e_F(i,3) + e_F(i,4)\big) - 5, \qquad i = 0, 1.$$

For each $i = 0, 1$*, the vector*

$$k_i(F) = 5\sum_{j=3}^{4} e_F(i,j)e_F(2,j) - 2\sum_{j=3}^{4} e_F(2,j) \tag{31}$$

lies in the rational span of h_0, h_1 *and the restricted strictly positive square vectors of this type.*

Proof. The same witness file gives two explicit combinations, each with 30 nonzero coefficients. Their available sources are the 56 positive edge-type square vectors and the two degree rows h_0, h_1. The checker computes those rows and the targets (31) from the flag edge sets, then verifies both identities in all 32 triangle-free coordinates. These are consequences of the single 1044-host certificate; no separate triangle-free extremal certificate is required. □

5. The sharp inequality

Proof of the inequalities in Theorems 1.1 and 1.3. Let M be the common supremum supplied by Lemma 2.2, and choose a global graphon maximum attaining it. The Clebsch construction gives $M \geq R_*$. Proposition 3.3 shows that its active probability edge density is strictly greater than $p_* = 5/144$. Lemma 2.5 therefore produces a global maximum at edge density p_*. All vertex, edge, and profile conditions used in the exact certificate hold at this graphon.

At fixed density p_*, a ratio at least R_* is equivalent to $\theta \geq \tau$. Corollary 4.2 gives $\theta \leq \tau$, so $M = R_*$. By the definition of M, the bound holds for every positive-density graphon and for every finite nonnegative weighted graph. If the edge mass is zero, the induced pentagon mass is zero as well. The Clebsch construction verifies sharpness. □

6. Eliminating triangles at equality

The equality witness from Lemma 4.3 has a direct consequence for the underlying graphon. Its use requires only the positive path-type squares and first-order vertex variation. We work on a probability space in this section.

Proposition 6.1. *Let W be a global maximizer of $C(W)/E(W)^{5/2}$ among graphons of positive edge density, where $C(W) = t_{\mathrm{ind}}(C_5, W)/10$ and $E(W) = t(K_2, W)/2$. If*

$$p(W) = p_* := \frac{5}{144}, \qquad t_{\mathrm{ind}}(C_5, W) = \tau := \frac{5}{663552},$$

then $t(K_3, W) = 0$.

Proof. Write $d(x) = \int W(x, y)\, dy$ and $\kappa = 5C(W)/(2E(W)) = \tau/(2p_*) > 0$. Let $L(x)$ denote one half of the rooted induced-pentagon density at x. Bounded signed variations of the vertex measure, followed by normalization to total mass one, give

$$L(x) = \kappa d(x) \qquad \text{for almost every } x. \tag{32}$$

Indeed, the normalization cancels from the homogeneous ratio, and its derivative in a bounded direction a is a positive constant times $\int a(x)(L(x) - \kappa d(x))\, dx$.

At the stated values of p and t, the verified certificate is an equality. Every strictly positively weighted square in its path-type block therefore vanishes. For ordered root locations x_0, x_1, x_2, put

$$\rho(x_0, x_1, x_2) = W(x_0, x_1)W(x_1, x_2)(1 - W(x_0, x_2)),$$

and let $\phi(x_0, x_1, x_2)$ be the conditional vector of the 72 five-vertex flags of this rooted type, with the root-type factor omitted. The entries of ϕ are nonnegative and sum to one. If v_j is one of the positive square vectors, then

$$\int \rho\,(v_j \cdot \phi)^2 = 0.$$

Thus all $v_j \cdot \phi$ vanish for $\rho(x_0, x_1, x_2)\,dx_0\,dx_1\,dx_2$-almost every root triple. Lemma 4.3 supplies a rational linear combination w of these vectors which is zero on the triangle-free flags and at least one on every flag containing a triangle. Consequently

$$0 \le \sum_{F \text{ containing a triangle}} \phi_F \le w \cdot \phi = 0$$

on the same set of root triples.

Define

$$T(x) = \int W(x, v)W(x, w)W(v, w)\,dv\,dw,$$
$$r(x) = \int W(x, y)W(y, z)(1 - W(x, z))\,dy\,dz.$$

Among the forbidden flag extensions is the event that the endpoint root x_0 and the two extension vertices span a triangle. Its conditional probability is exactly $T(x_0)$: edges from the extensions to roots 1 and 2 are unrestricted and are summed out. Hence

$$0 = \int \rho(x_0, x_1, x_2)T(x_0)\,dx_0\,dx_1\,dx_2 = \int r(x)T(x)\,dx. \tag{33}$$

In the rooted induced cycle $x - a - b - c - d - x$, retain only the factors $W(x, a)W(a, b)(1 - W(x, b))$. All other factors lie in $[0, 1]$, and the two remaining variables integrate to one. The factor $1/2$ in the definition of L therefore gives

$$L(x) \le \frac{r(x)}{2}.$$

Together with (32), this yields $r(x) \ge 2\kappa d(x)$. Since every term is nonnegative, (33) implies $\int d(x)T(x)\,dx = 0$. Finally $0 \le T(x) \le d(x)^2$. Thus T vanishes both where $d = 0$ and where $d > 0$, up to null sets. It follows that $t(K_3, W) = \int T(x)\,dx = 0$. □

7. Classification of equality

The preceding triangle-elimination argument reduces equality in the unrestricted problem to a triangle-free graphon at the reference density. We now restrict the positive-square vectors of the same certificate to triangle-free flags and use the two exact conditional spans recorded in Section 4.

7.1. **Structure at the reference density.**

Proposition 7.1. *Let W be a triangle-free global ratio maximizer with $(p, \theta) = (p_*, \tau)$. Then W is a balanced Clebsch graphon on an active set of measure $1/3$, with isolates outside it.*

Proof. By Corollary 4.2, every strictly positive square in the exact certificate vanishes. For each such square vector v, formula (26) gives $v \cdot q(\mathbf{x}) = 0$ for almost every root tuple with respect to its type measure.

First use the path type $01, 12$. The conditional expectation of the vector in Lemma 4.4 is $48d(x_1) - 5$. Its type probability is

$$W(x_0, x_1)W(x_1, x_2)(1 - W(x_0, x_2)).$$

The last factor is redundant almost everywhere on the product of the first two, by triangle-freeness. Integrating over both endpoints therefore gives

$$\int d(x)^2(48d(x) - 5)^2 \, dx = 0.$$

Thus

$$d(x) \in \left\{0, \frac{5}{48}\right\} \qquad \text{almost everywhere.} \tag{34}$$

Let $A = \{x : d(x) > 0\}$. Its complement has zero rows almost everywhere, and

$$|A| = \frac{p_*}{5/48} = \frac{1}{3}.$$

Next consider the type consisting of the edge 01 and isolated root 2. Its two edge endpoints have positive degree for almost every root tuple, so the conditional expectations of h_0, h_1 in Lemma 4.5 vanish by (34). The same lemma then implies

$$c(x_i, x_2) = \frac{2}{5} d(x_2), \qquad i = 0, 1, \tag{35}$$

almost everywhere with respect to this type measure. Indeed the conditional expectation of k_i is $10c(x_i, x_2) - 4d(x_2)$.

Restrict to A and normalize its measure to one. Call the resulting graphon U. Its degree is $p_0 = 5/16$. For type tuples entirely in A, (35) gives codegree $\mu = 1/8$ in the new normalization. We now show that these relations determine U.

Write

$$c_U(x, z) = \int U(x, y)U(z, y) \, dy.$$

Triangle-freeness gives $U(x, z)c_U(x, z) = 0$ almost everywhere. Integrate the square of $c_U(x, z) - \mu$ against the edge-plus-isolate type probability and integrate out its middle root. Since

$$\int U(x, y)(1 - U(y, z)) \, dy = p_0 - c_U(x, z),$$

we obtain

$$(1 - U(x,z))(p_0 - c_U(x,z))(c_U(x,z) - \mu)^2 = 0 \quad \text{almost everywhere.} \tag{36}$$

If $0 < U(x,z) < 1$, triangle-freeness forces $c_U(x,z) = 0$, contradicting (36). Hence U is $\{0,1\}$-valued. Edges have codegree zero, and nonedges have codegree μ or p_0.

Regard $f_x = U(x,\cdot)$ as an element of $L^2([0,1])$. After a null set is discarded, all rows are $\{0,1\}$-valued and have integral p_0. Therefore

$$\|f_x - f_z\|_2^2 = 2p_0 - 2c_U(x,z).$$

In particular $c_U(x,z) = p_0$ precisely when the two rows agree in L^2. Each twin class $T_x = \{z : f_z = f_x\}$ is measurable: it is the zero set of the measurable function $z \mapsto \int |U(z,y) - U(x,y)|^2 \, dy$.

Let $a(x) = |T_x|$. Almost every twin is a nonneighbor, and every other nonneighbor has codegree μ. Regularity gives $\int c_U(x,z)\, dz = p_0^2$, whence

$$p_0^2 = a(x)p_0 + (1 - p_0 - a(x))\mu.$$

Substituting $p_0 = 5/16$ and $\mu = 1/8$ gives $a(x) = 1/16$ for almost every x. Choose generic representatives successively outside the classes already selected. Since each selected class has measure $1/16$, exactly sixteen disjoint measurable classes cover the space up to null sets.

The kernel is constant on every class rectangle. To see this, on a twin class A_i all rows agree in L^2 with one function f_i. Fubini and symmetry give $U(x,z) = f_i(z) = f_j(x)$ almost everywhere on $A_i \times A_j$. Integrating over either coordinate makes this value constant. Every block value is zero or one, and diagonal blocks are zero because their pairs have positive codegree. The quotient is consequently a simple graph on sixteen vertices, of degree five, with zero common neighbors for adjacent vertices and two for distinct nonadjacent vertices.

These parameters determine the Clebsch graph elementarily. Fix a vertex v with independent neighbors $u_1, \ldots, u_5$. Every other vertex has two neighbors among these five. Each pair u_i, u_j has exactly two common neighbors, one being v; label the other w_{ij}. This labels the remaining ten vertices by two-element subsets of $\{1, \ldots, 5\}$. Two pair vertices sharing a label cannot be adjacent, since that would create a triangle. A pair vertex needs three further neighbors after u_i, u_j, and there are exactly three disjoint pairs. It joins precisely those three.

Choose the five Clebsch generators $s_1, \ldots, s_5$ from the introduction. The assignment

$$v \mapsto 0, \qquad u_i \mapsto s_i, \qquad w_{ij} \mapsto s_i + s_j$$

is an isomorphism to the Clebsch graph: two pair sums are adjacent exactly when their pairs are disjoint. Thus U is balanced Clebsch, proving the proposition. □

7.2. **Arbitrary equality graphons.**

Equality in Theorem 1.3. Every member of $\mathcal{E}_{\mathrm{Cl}}$ attains the bound by the Clebsch counts and homogeneity. Conversely, suppose $R(W) = R_*$ and $E(W) > 0$. The sharp inequality has already been proved for all graphons, so W is a global maximum over all positive edge densities. Remove its actual zero-degree set and normalize the remaining vertex measure to one; the ratio and global maximality are unchanged.

Proposition 3.3 applies to this graphon even if its values are fractional. Its active density is therefore greater than p_*. Pad by isolates as in Lemma 2.5. The padded graphon is still a global maximum, has $p = p_*$ and $\theta = \tau$, and satisfies the vertex first-order conditions. Proposition 6.1 makes it triangle-free. Proposition 7.1 then identifies its active part as a balanced Clebsch graphon.

Removing the added isolates and undoing the common vertex-measure scaling proves $W \in \mathcal{E}_{\mathrm{Cl}}$. This argument classifies an arbitrary equality graphon. It does not infer graphon equality from equality on finite graphs, and it does not impose edge optimality on a merely template-optimal finite weighting. □

Equality in Theorem 1.1. Normalize the total positive vertex weight to one and represent the weighted graph as a step graphon with those class measures. At equality, Theorem 1.3 gives a balanced Clebsch active part and isolates. Each original positive-weight vertex has a constant neighborhood row, so its cell belongs to one of the twin classes, or to the isolate set, up to null sets. The sixteen active classes therefore have equal total vertex weight, and their adjacencies are exactly the Clebsch adjacencies. The converse follows from the counts $E = 40a^2$ and $C = 192a^5$ when each active class has total weight a. □

8. Stability on the edge scale

The number of isolated or low-degree vertices is not controlled by an edge constraint. We first remove a small amount of edge mass to obtain a core with positive normalized minimum degree. Compactness then places that core in the neighborhood covered by the local stability theorem. A final rounding argument produces equal integer class sizes without an additive error term.

Theorem 8.1 (Edge-scaled stability)**.** *There are absolute constants $K < \infty$ and $\varepsilon_0 > 0$ such that the following holds. Let G be a finite graph with $m > 0$ edges, and suppose that $0 \le \varepsilon \le \varepsilon_0$ and*

$$I_5(G) \ge (R_* - \varepsilon)\left(\frac{m}{5}\right)^{5/2}.$$

There is a graph B on the same vertex set, consisting of a Clebsch blow-up with sixteen classes of the same positive integer size and isolated vertices, for

which

$$|E(G) \triangle E(B)| \le K\sqrt{\varepsilon}\, m.$$

If s is the common class size, then

$$|m - 40s^2| \le K\sqrt{\varepsilon}\, m, \qquad s = (1 + O(\sqrt{\varepsilon}))\sqrt{\frac{m}{40}}.$$

There is no restriction on $|V(G)|$ and no additional rounding term. The square-root dependence on the deficit is optimal, as shown in Proposition 8.5 below.

8.1. **Removing low-degree vertices.**

Lemma 8.2 (Weighted peeling)**.** *Let (H, x) be a finite weighted graph with $E(H, x) = 1$ and $C(H, x) \ge c_* - \eta$, where $0 \le \eta \le c_*/4$. Repeatedly remove a vertex whose current weighted degree is at most c_*, keeping the other weights fixed. Write (H', x') for the remaining core, $E' = E(H', x')$, $C' = C(H', x')$, and $W' = \sum_{v \in V(H')} x'_v$. Then*

$$1 - E' \le \frac{4\eta}{3c_*}, \tag{37}$$

$$\frac{C'}{(E')^{5/2}} \ge C(H, x), \tag{38}$$

$$W' < \frac{2}{c_*}. \tag{39}$$

In particular $E' > 0$. After the core weights are normalized to have sum one, its weighted degrees are all at least $c_*^2/2$.

Proof. At an intermediate stage, let E_t be the current edge mass and let d_i be the current weighted degree of a vertex i about to be removed. Fixing the edge opposite i in a pentagon gives

$$C_i \le \frac{1}{4} E_t d_i^2.$$

Indeed, for a fixed opposite edge uv, the two neighbors of i lie in the exclusive u-neighbor and exclusive v-neighbor subsets of $N(i)$. Inducedness makes these sets disjoint even when the graph has triangles. Their weight product is at most $d_i^2/4$. Removing i destroys edge mass $x_i d_i$ and pentagon mass $x_i C_i$. As $E_t \le 1$ and $d_i \le c_*$, the latter is at most $(c_*/4) x_i d_i$.

Put $\alpha = 1 - E'$. Each removed edge and each destroyed pentagon is charged at its first deleted endpoint, so

$$0 \le C(H, x) - C' \le \frac{c_*}{4}\alpha. \tag{40}$$

Theorem 1.1, applied to the surviving core, gives

$$c_* - \eta - \frac{c_*}{4}\alpha \le C' \le c_*(1 - \alpha)^{5/2}.$$

Since $1-(1-\alpha)^{5/2} \geq \alpha$ for $0 \leq \alpha \leq 1$, this proves (37), and hence $E' \geq 2/3$. Moreover $C(H,x) \geq 3c_*/4$, so

$$C' \geq C(H,x) - \frac{c_*}{4}\alpha \geq C(H,x)(1-\alpha) \geq C(H,x)(1-\alpha)^{5/2}.$$

This is (38).

Every surviving vertex has weighted degree greater than c_*. Consequently

$$c_* W' < \sum_{i \in V(H')} x_i d_i = 2E' \leq 2.$$

After dividing the vertex weights by W', every weighted degree exceeds

$$\frac{c_*}{W'} > \frac{c_*^2}{2E'} \geq \frac{c_*^2}{2}.$$

The degrees must be recomputed after each removal; this is why the procedure is iterative. □

8.2. **Entering the local neighborhood.** Let W_{Cl} denote a full balanced Clebsch graphon. We use $\delta_\square$ for cut distance and δ_1 for L^1 distance, in each case taking the infimum over measure-preserving relabelings. The standard compactness of graphons in cut distance and continuity of fixed finite homomorphism densities will be used only qualitatively.

Lemma 8.3. *Fix $\mu > 0$. If W_j are graphons with $d_{W_j} \geq \mu$ almost everywhere and $R(W_j) \to R_*$, then $\delta_1(W_j, W_{\mathrm{Cl}}) \to 0$.*

Proof. Their edge densities are at least μ. From any subsequence, compactness extracts a further subsequence converging in cut distance to a graphon W. Choose relabelings giving cut-norm convergence. For every measurable $S \subseteq [0,1]$,

$$\int_S d_W = \lim_j \int_S d_{W_j} \geq \mu|S|,$$

because the difference of the two integrals is bounded by the cut norm. Thus $d_W \geq \mu$ almost everywhere. The equality $R(W) = R_*$ passes to the limit by induced-density continuity; the denominator in R stays bounded away from zero. Theorem 1.3 identifies W as balanced Clebsch on its active set, with isolates outside. The minimum-degree condition excludes a positive-measure isolate set, so W is a full balanced Clebsch graphon.

This proves convergence of the whole sequence in cut distance. To obtain L^1 convergence, fix the sixteen classes $A_1, \ldots, A_{16}$ of a labeling of W_{Cl}. Since W_{Cl} is $\{0,1\}$-valued, $U - W_{\mathrm{Cl}}$ has a constant sign on each $A_i \times A_j$. Hence

$$\|U - W_{\mathrm{Cl}}\|_1 = \sum_{i,j} \left| \int_{A_i \times A_j} (U - W_{\mathrm{Cl}}) \right| \leq 256 \|U - W_{\mathrm{Cl}}\|_\square.$$

Taking infima over relabelings completes the proof. □

8.3. **Rounding whole vertices.** Write $\mathcal{E}_{\mathrm{Cl}}$ for the family defined in the introduction. The next lemma is useful because balancing real class masses by rounding their sizes separately would introduce an unnecessary additive error.

Lemma 8.4 (Integral rounding)**.** *Let W_G be the probability graphon of an n-vertex graph, and let $U \in \mathcal{E}_{\mathrm{Cl}}$ have active mass $q \geq 1/2$. There is a graph B on $V(G)$, consisting of sixteen equal integer Clebsch classes and isolates, such that*

$$\|W_G - W_B\|_1 \leq 897\|W_G - U\|_1.$$

The common class size is positive when $\|W_G - U\|_1$ is sufficiently small.

Proof. Put $\zeta = \|W_G - U\|_1$. Let B_0 be the isolate set of U, and let $B_1, \dots, B_{16}$ be its active classes, each of measure $b = q/16$. Denote the corresponding row functions by $r_0 = 0, r_1, \dots, r_{16}$. Different rows have distance at least

$$\rho = \frac{5q}{16} \geq \frac{5}{32}. \tag{41}$$

A nonzero row meets five classes, while distinct Clebsch rows differ on six or ten classes.

For a vertex v, let I_v be its cell in W_G, of measure $1/n$, and let g_v be its constant row. Set

$$p_{v,i} = n|I_v \cap B_i|, \qquad e_{v,i} = \|g_v - r_i\|_1, \qquad e_v = \sum_{i=0}^{16} p_{v,i} e_{v,i}.$$

Assign the entire vertex v to a type $i(v)$ minimizing $e_{v,i}$. If this minimum is at least $\rho/2$, then $e_v \geq \rho/2$. Otherwise, the triangle inequality and (41) give $e_{v,j} \geq \rho/2$ for every $j \neq i(v)$. In both cases

$$1 - p_{v,i(v)} \leq \frac{2e_v}{\rho}.$$

The total measure whose type changes is therefore

$$M = \frac{1}{n}\sum_v (1 - p_{v,i(v)}) \leq \frac{2\zeta}{\rho}. \tag{42}$$

Let T be the Clebsch step graphon given by these whole-vertex assignments, and let s_i be the mass of its ith active class. Changing types on mass M gives

$$\|T - U\|_1 \leq 2M, \qquad \sum_{i=1}^{16} |s_i - b| \leq 2M.$$

Set $s_{\min} = \min_i s_i$. Remove excess vertices from each active class until every class has size $ns_{\min}$, assigning the removed vertices to the isolate type. This uses whole vertices because all ns_i are integers. The total mass reassigned is

$$\Delta = \sum_{i=1}^{16} (s_i - s_{\min}) \leq 17 \sum_{i=1}^{16} |s_i - b| \leq 34M.$$

For the first inequality, if $b \geq s_{\min}$ use $b - s_{\min} \leq \sum_i |s_i - b|$; if $b < s_{\min}$, the inequality is immediate. The additional L^1 cost is at most $2\Delta \leq 68M$. The resulting graph B is exactly balanced, and

$$\|W_G - W_B\|_1 \leq \zeta + 2M + 68M \leq \left(1 + \frac{140}{\rho}\right)\zeta \leq 897\zeta.$$

For sufficiently small ζ, (42) and $b \geq 1/32$ also imply $s_i > 0$ for every i. □

This proof applies after arbitrary relabelings: the cells I_v need not be intervals. It also handles a nonattained distance infimum. Apply the lemma to a sequence of comparison graphons approaching the infimum and use the finiteness of the possible edit distances on a fixed vertex set. If the infimum is zero, taking the error small enough to permit fewer than one edge edit forces an exact copy.

8.4. **Proof of the stability theorem.**

Proof of Theorem 8.1. Give every vertex of G weight $1/\sqrt{m}$. Then $E = 1$ and

$$C \geq c_* - \eta, \qquad \eta = \frac{\varepsilon}{5^{5/2}}.$$

Apply Lemma 8.2. The peeling removes at most

$$\frac{4\varepsilon}{3c_* 5^{5/2}} m \tag{43}$$

edges and leaves a graph H with

$$n_H < \frac{2\sqrt{m}}{c_*}.$$

Its probability graphon has minimum degree at least $c_*^2/2$ and ratio at least $R_* - \varepsilon$.

By Lemma 8.3, there is an absolute positive threshold for ε below which W_H, after a relabeling, lies in the neighborhood of Theorem A.1. Let $c_{\mathrm{loc}} > 0$ be the constant in that theorem. Its induced-pentagon functional is exactly the functional used here, so the local conclusion is

$$d_1(W_H, \mathcal{E}_{\mathrm{Cl}}) \leq \sqrt{\frac{\varepsilon}{c_{\mathrm{loc}}}}.$$

Comparison graphons sufficiently close to this infimum have active mass at least $1/2$ when ε is small. To see this, also choose the compactness neighborhood small enough that W_H is close to the full Clebsch graphon in L^1. Edge-density continuity and $t(K_2, U) = (5/16)q^2$ then force q close to one.

Lemma 8.4 gives a balanced Clebsch blow-up B_H with isolates on $V(H)$. Taking the comparison distance to its infimum, the number of edits inside H is at most

$$\frac{897 n_H^2}{2}\sqrt{\frac{\varepsilon}{c_{\mathrm{loc}}}} \leq \frac{1794}{c_*^2\sqrt{c_{\mathrm{loc}}}}\sqrt{\varepsilon}\, m.$$

Extend B_H to the original vertex set by making every peeled vertex isolated. Adding (43), and using $\varepsilon \le \sqrt{\varepsilon}$ for $\varepsilon \le 1$, proves the edit bound. For example, once ε_0 has been chosen small enough for the preceding neighborhood and nonemptiness requirements, one may take

$$K = \frac{1794}{c_*^2\sqrt{c_{\mathrm{loc}}}} + \frac{4}{3c_*5^{5/2}}.$$

The value of K is not optimized. Finally, a balanced Clebsch blow-up with common class size s has $40s^2$ edges. Its edge count differs from m by at most the number of edits, proving both remaining conclusions. □

8.5. **Optimality of the exponent.**

Proposition 8.5. *The factor $\sqrt{\varepsilon}$ in Theorem 8.1 cannot be replaced by $o(\sqrt{\varepsilon})$, uniformly over finite graphs.*

Proof. Choose two nonadjacent vertices u, v of Cl. For integers $k \ge 2$ and $1 \le r < k$, form a Clebsch blow-up $G_{k,r}$ with class sizes $k + r$ at u, $k - r$ at v, and k at the other fourteen vertices.

Each Clebsch vertex lies in sixty pentagons, and each nonadjacent pair lies in twelve. For the latter count, the pair has two common neighbors and six three-edge paths between its endpoints. Each choice of a common neighbor and such a path forms a pentagon, by triangle-freeness. The six paths can also be counted from the five neighbors of one endpoint: two are adjacent to the other endpoint and contribute zero codegree, while the other three contribute codegree two.

The edge and pentagon polynomials are affine in each class size. The linear changes at u and v cancel, and there is no quadratic edge term because u, v are nonadjacent. Thus, exactly,

$$|V(G_{k,r})| = 16k, \qquad m = 40k^2, \qquad I_5(G_{k,r}) = 192k^5 - 12k^3r^2.$$

Its normalized deficit is

$$\varepsilon_{k,r} = R_* - \frac{I_5(G_{k,r})}{(m/5)^{5/2}} = \frac{R_* r^2}{16k^2}.$$

The pair u, v has two common neighbors. Its three exclusive neighbors on each side have unchanged class size k, so $G_{k,r}$ has $3k$ vertices of degree $5k+r$, $3k$ of degree $5k - r$, and $10k$ of degree $5k$.

A balanced Clebsch blow-up with isolates on the same $16k$ vertices has common class size at most k, and hence maximum degree at most $5k$. Each of the $3k$ high-degree vertices must lose at least r in degree. One edge edit changes the total absolute degree discrepancy by at most two. Therefore every such target B satisfies

$$|E(G_{k,r}) \triangle E(B)| \ge \frac{3}{2}kr = \frac{3}{20\sqrt{R_*}}\sqrt{\varepsilon_{k,r}}\, m.$$

Taking $r = 1$ and $k \to \infty$ proves the assertion. □

9. Linear structural stability and quadratic mass stability

The local repair argument proves a stronger conclusion than its quadratic distance estimate alone. Edge errors have a linear cost in the extremal deficit; only the imbalance of the sixteen class masses has a quadratic cost. We retain these two conclusions separately.

Let $\mathcal{T}_{\mathrm{Cl}}$ denote the class of graphons obtained from an arbitrary sixteen-class Clebsch blow-up and an isolated class. The sixteen active masses need not be equal. For a presented member T with active masses $s_1, \dots, s_{16}$, put

$$q(T) = \sum_{i=1}^{16} s_i, \qquad V(T)^2 = \sum_{i=1}^{16} (s_i - q(T)/16)^2.$$

Proposition 9.1 (The two local defects)**.** *There are constants $A, B > 0$ and $\delta > 0$ such that, whenever $\|W - W_{\mathrm{Cl}}\|_1 < \delta$, a graphon $T \in \mathcal{T}_{\mathrm{Cl}}$ exists with all active masses at least $1/32$ and*

$$\|W - T\|_1 \le AD(W), \qquad V(T)^2 \le BD(W), \qquad D(W) = R_* - R(W). \tag{44}$$

Proof. Use the sets $B_0, Z, M_1, \dots, M_{16}$ constructed in the proof of Theorem A.1; here B_0 denotes its bad-row set to avoid confusion with the constant in the present statement. Write $F = C/E^{5/2} = R/5^{5/2}$ as in that proof. Take T to be the repaired Clebsch step graphon on $M = \bigcup_i M_i$, and put isolates outside M. Before balancing the masses, the four-defect inequality is

$$F(W_{\mathrm{Cl}}) - F(W) \ge c_B \mu_{B_0} + c_Z D_Z + c_E e_M + c_H V(T)^2.$$

The three operations that produce T cost at most $2\mu_{B_0} + 2D_Z + e_M$ in L^1. Consequently one may take

$$A = \frac{1}{5^{5/2} \min\{c_B/2, c_Z/2, c_E\}}, \qquad B = \frac{1}{5^{5/2} c_H}.$$

The mass-closeness estimate in that proof gives $s_i = 1/16 + O(\|W - W_{\mathrm{Cl}}\|_1)$ uniformly in i. Decreasing the fixed neighborhood ensures $s_i \ge 1/32$. □

Lemma 9.2 (Rounding without balancing)**.** *Let W_G be the probability graphon of a graph on n vertices, and let $U \in \mathcal{T}_{\mathrm{Cl}}$ have active masses $s_i \ge 1/32$. Set $\zeta = \|W_G - U\|_1$ and $\rho = 5/32$. There is a Clebsch blow-up H, with isolates, on the same vertex set, such that*

$$\|W_G - W_H\|_1 \le (1 + 4/\rho)\zeta, \tag{45}$$

$$V(W_H)^2 \le 2V(U)^2 + 32\zeta^2/\rho^2. \tag{46}$$

All sixteen active classes of H are nonempty if $\zeta < \rho/128$.

Proof. The seventeen row prototypes of U are pairwise separated in L^1 by at least ρ. Indeed, a nonzero Clebsch row meets five classes, while distinct nonzero rows differ on at least six classes.

For each vertex cell I_v, write $p_{v,i} = n|I_v \cap A_i|$ and $e_{v,i} = \|W_G(v, \cdot) - r_i\|_1$, including the isolated type $i = 0$. Assign the whole vertex to a type

minimizing $e_{v,i}$. The argument of Lemma 8.4 gives total changed type mass

$$M = \frac{1}{n}\sum_{v}(1 - p_{v,i(v)}) \le \frac{2\zeta}{\rho}.$$

The induced whole-vertex template W_H differs from U only on pairs with at least one endpoint in this changed set. Therefore $\|U - W_H\|_1 \le 2M$, proving (45). No trimming is performed.

Let s' be the vector of the sixteen rounded active masses. The changed mass estimate implies $\|s' - s\|_1 \le 2M$. Let P be orthogonal projection onto the sum-zero subspace of $\mathbb{R}^{16}$. Then

$$\|Ps'\|_2^2 \le 2\|Ps\|_2^2 + 2\|P(s' - s)\|_2^2 \le 2V(U)^2 + 8M^2,$$

which is (46). Finally $s'_i \ge s_i - 2M \ge 1/32 - 4\zeta/\rho > 0$ under the stated smallness assumption. Each ns'_i is an integer because only whole vertices were assigned. The vertex cells can be arbitrary measurable sets, so the argument also applies after a graphon relabeling. □

Theorem 9.3 (Linear structural stability)**.** *There are absolute constants $K < \infty$ and $\varepsilon_0 > 0$ such that the following holds. Suppose G has $m > 0$ edges and*

$$I_5(G) \ge (R_* - \varepsilon)(m/5)^{5/2}, \qquad 0 \le \varepsilon \le \varepsilon_0.$$

Then a graph H on the same vertex set, consisting of a Clebsch blow-up with positive integer class sizes $n_1, \dots, n_{16}$ and isolated vertices, satisfies

$$|E(G) \triangle E(H)| + \sum_{i=1}^{16}(n_i - N/16)^2 \le K\varepsilon m, \qquad N = \sum_{i=1}^{16} n_i. \tag{47}$$

In particular,

$$\left| m - \frac{5N^2}{32} \right| \le K'\varepsilon m, \qquad N = \sqrt{\frac{32m}{5}} + O(\varepsilon\sqrt{m}), \tag{48}$$

and every class satisfies

$$n_i = (1 + O(\sqrt{\varepsilon}))\sqrt{m/40}.$$

All constants are absolute. There is no additive rounding term.

Proof. Give every vertex weight $1/\sqrt{m}$. The weighted peeling lemma removes at most $4\varepsilon m/(3R_*)$ edges, and leaves a core G' with $n < 2\sqrt{m}/c_*$ vertices, normalized minimum degree at least $c_*^2/2$, and ratio at least $R_* - \varepsilon$. By the positive-degree compactness lemma, after relabeling its probability graphon lies in the neighborhood of Proposition 9.1 when ε_0 is small enough. Obtain the template U there. It satisfies

$$\zeta \le A\varepsilon, \qquad V(U)^2 \le B\varepsilon.$$

Decrease ε_0 further so that $A\varepsilon_0 < \rho/128$ and $\varepsilon_0 \le 1$. Lemma 9.2 gives whole-vertex classes with

$$\|W_{G'} - W_H\|_1 \le (1 + 4/\rho)A\varepsilon,$$
$$V(W_H)^2 \le B'\varepsilon, \qquad B' = 2B + 32A^2/\rho^2.$$

The number of edge edits inside the core is $n^2/2$ times its L^1 distance. Also $\sum_i (n_i - N/16)^2 = n^2 V(W_H)^2$. Using $n^2 < 4m/c_*^2$ and putting all peeled vertices into the isolated class proves the two estimates separately with constants

$$K_E = \frac{4}{3R_*} + \frac{2(1 + 4/\rho)A}{c_*^2}, \qquad K_V = \frac{4B'}{c_*^2}.$$

Take $K = K_E + K_V$ in (47).

For completeness, let A_{Cl} be the Clebsch adjacency matrix and $h_i = n_i - N/16$. Since the graph is five-regular and $\sum_i h_i = 0$,

$$e(H) = 40(N/16)^2 + \frac{1}{2}h^T A_{\mathrm{Cl}} h = \frac{5N^2}{32} + \frac{1}{2}h^T A_{\mathrm{Cl}} h.$$

The elementary bound $|h^T A_{\mathrm{Cl}} h| \le 5\|h\|_2^2$ follows from $2|h_i h_j| \le h_i^2 + h_j^2$ summed over the edges. Hence one may take $K' = K_E + (5/2)K_V$ in the first estimate of (48). For smaller ε_0, this places N between fixed positive multiples of $\sqrt{m}$. Factoring the difference of squares yields the second estimate. Finally $|h_i| \le \|h\|_2 = O(\sqrt{\varepsilon m})$, which gives the class-size conclusion. □

Proposition 9.4 (Optimality of linear structural stability)**.** *The factor ε in the edit-distance conclusion of Theorem 9.3 cannot be replaced uniformly by $o(\varepsilon)$.*

Proof. Start with a balanced Clebsch blow-up of common class size $k \ge 2$, and add a single edge between two vertices in one class. The two endpoints are now true twins. An induced pentagon cannot contain both, and all other induced pentagons are unchanged. Thus the new graph has

$$m_k = 40k^2 + 1, \qquad I_5 = 192k^5, \qquad \varepsilon_k = R_* \left[1 - \left(\frac{40k^2}{40k^2 + 1}\right)^{5/2}\right].$$

Its distance to the family of Clebsch blow-ups and isolates is exactly one: deleting the added edge suffices, and the graph contains a triangle whereas every graph in that family is triangle-free. Furthermore $\varepsilon_k \to 0$ and $\varepsilon_k m_k \to (5/2)R_*$. A bound of the form $o(\varepsilon_k)m_k$ would tend to zero, contradicting this integer distance of one. □

9.1. **A two-sided geometric description of the deficit.** For a graph G with $m > 0$ edges, write

$$\Delta(G) = c_* m^{5/2} - I_5(G).$$

Let $\mathcal{B}_m(G)$ consist of the Clebsch blow-ups with isolates on $V(G)$ whose sixteen class sizes obey

$$\frac{1}{2}\sqrt{m/40} \le n_i \le 2\sqrt{m/40}.$$

For such a template put $N = \sum_i n_i$ and define

$$J_G(B) = |E(G) \triangle E(B)| + \sum_i (n_i - N/16)^2.$$

Proposition 9.5 (Two-sided structural energy)**.** *For absolute constants $c, C > 0$, every graph sufficiently close to the sharp ratio satisfies*

$$c\, m^{3/2} \min_{B \in \mathcal{B}_m(G)} J_G(B) \le \Delta(G) \le C\, m^{3/2} \min_{B \in \mathcal{B}_m(G)} J_G(B). \tag{49}$$

In particular $\mathcal{B}_m(G)$ is nonempty in this range. The upper estimate holds whenever $\mathcal{B}_m(G)$ is nonempty, without a separate near-extremality assumption.

Proof. For a sufficiently small fixed normalized deficit, the class-size conclusion of Theorem 9.3 places its template in $\mathcal{B}_m(G)$. Substituting $\varepsilon = 5^{5/2}\Delta(G)/m^{5/2}$ in that theorem proves the lower estimate in (49).

For the other direction, fix $B \in \mathcal{B}_m(G)$ and put

$$t = |E(G) \triangle E(B)|, \qquad V = \sum_i (n_i - N/16)^2, \qquad m_B = e(B).$$

An original induced pentagon of B is still induced in G unless one of its ten pairs is modified. Each modified pair belongs to at most $\binom{N-2}{3} \le N^3/6$ original five-sets; pairs meeting an isolated vertex of B belong to none. Thus

$$I_5(G) \ge I_5(B) - tN^3/6. \tag{50}$$

This estimate depends on the number of active template vertices, not on the possibly much larger number of vertices of G.

Consider the weighted-template defect

$$Q(x) = c_* E(x)^{5/2} - P(x).$$

At every positive constant vector $b\mathbf{1}$ we have $Q = 0$ and $\nabla Q = 0$. The latter follows either by differentiating the sharp equality along class weights, or directly from $E(\mathbf{1}) = 40$, $P(\mathbf{1}) = 192$, $\nabla E(\mathbf{1}) = 5\mathbf{1}$ and $\nabla P(\mathbf{1}) = 60\mathbf{1}$. Set $b = N/16$ and $h_i = n_i - b$. Every point of the line segment $b\mathbf{1} + uh$, $0 \le u \le 1$, has coordinates between fixed positive multiples of $\sqrt{m}$. On this segment the Hessian of Q is $O(m^{3/2})$: the polynomial P has degree five, and the same scaling follows directly by differentiating $E^{5/2}$, with E bounded above and below by fixed positive multiples of m. Taylor's formula therefore gives

$$0 \le c_* m_B^{5/2} - I_5(B) = Q(b\mathbf{1} + h) \le C_0 m^{3/2} \|h\|_2^2 = C_0 m^{3/2} V.$$

Also m_B and m are within fixed positive constant factors, and $|m_B - m| \le t$, so

$$|m^{5/2} - m_B^{5/2}| \le C_1 m^{3/2} t.$$

Since $N = O(\sqrt{m})$, combine these estimates with (50) to obtain $\Delta(G) \le Cm^{3/2}(t+V)$. Taking the minimum completes the proof. There are finitely many vertex partitions, so the minimum is attained whenever the family is nonempty. □

10. The arithmetic size of the finite-edge deficit

The error term in Corollary 1.2 cannot be improved uniformly to $o(m^2)$. More precisely, its order is governed by the distance of an effective vertex count from the integers. This conclusion uses the linear stability theorem for unbalanced Clebsch blow-ups, including its squared class-size estimate.

For an integer $N \ge 0$, write $N = 16k + r$, with $0 \le r < 16$, and put

$$v(N) = \frac{r(16-r)}{16}, \qquad \Psi(m) = \min_{N \in \mathbb{Z}_{\ge 0}} \left(\left| m - \frac{5}{32} N^2 \right| + v(N) \right).$$

The minimum is attained, and $0 \le v(N) \le 4$. As before, let $\rho(C_5, m)$ denote the maximum number of induced pentagons among graphs with exactly m edges, and set

$$\Delta(m) = c_* m^{5/2} - \rho(C_5, m).$$

Theorem 10.1 (Arithmetic deficit law)**.** *There exist absolute constants $a_0, b_0 > 0$ and an integer m_0 such that, for every integer $m \ge m_0$,*

$$a_0 m^{3/2} \Psi(m) \le \Delta(m) \le b_0 m^{3/2} \Psi(m). \tag{51}$$

In particular, $\Psi(m) = 0$ exactly when $m = 40k^2$ for an integer $k \ge 0$. For all other sufficiently large integers m,

$$\Delta(m) \asymp m^2 \operatorname{dist}\left(\sqrt{\frac{32m}{5}}, \mathbb{Z} \right) + m^{3/2}, \tag{52}$$

where the two comparison constants are absolute.

Proof. We first record the role of the integer class sizes. If $n_1, \dots, n_{16}$ are nonnegative integers with sum N, then

$$V := \sum_{i=1}^{16} (n_i - N/16)^2 \ge v(N). \tag{53}$$

Indeed, whenever two entries differ by at least two, moving one unit from the larger to the smaller decreases the sum of their squares. The minimum is therefore attained by r entries equal to $k+1$ and $16-r$ entries equal to k; its value is $v(N)$.

Let A be the adjacency matrix of Cl, and write $n = (N/16)\mathbf{1} + h$, where $\sum_i h_i = 0$. Since $A\mathbf{1} = 5\mathbf{1}$, the edge count of this blow-up is

$$E(n) = \frac{1}{2} n^T A n = \frac{5}{32} N^2 + \frac{1}{2} h^T A h. \tag{54}$$

The operator norm of A is at most five. Consequently

$$\left|E(n) - \frac{5}{32}N^2\right| \le \frac{5}{2}V. \tag{55}$$

For the lower bound in (51), choose an extremal graph G with m edges. Such a graph exists, since discarding isolates leaves at most $2m$ vertices. The earlier lower construction gives $\Delta(m) = O(m^2)$. Hence its normalized deficit

$$\varepsilon = R_* - \frac{I_5(G)}{(m/5)^{5/2}} = \frac{5^{5/2}\Delta(m)}{m^{5/2}}$$

tends to zero as $m \to \infty$. Apply Theorem 9.3. It supplies a Clebsch blow-up H, together with isolates, on the same vertex set, such that its integer class sizes n_i, total active size N, squared deviation V, and edge-edit distance L satisfy

$$L + V \le K\varepsilon m = K5^{5/2}\frac{\Delta(m)}{m^{3/2}}.$$

Here and below a constant may be increased without changing its dependence. In particular $|m - E(n)| \le L$. Equations (53) and (55) give

$$\Psi(m) \le \left|m - \frac{5}{32}N^2\right| + v(N) \le L + \frac{7}{2}V \le K'\frac{\Delta(m)}{m^{3/2}}.$$

This proves the first inequality.

For the upper bound, put $t = \sqrt{32m/5}$. An integer N_0 nearest to t satisfies

$$\left|m - \frac{5}{32}N_0^2\right| \le \frac{5}{32}\left(t + \frac{1}{4}\right).$$

It follows that $\Psi(m) = O(\sqrt{m})$. Any minimizing N therefore satisfies $N = \Theta(\sqrt{m})$ and tends to infinity. Choose $n_1, \dots, n_{16}$ as equally as possible, so that $V = v(N)$, and let H be the resulting Clebsch blow-up. With $e = E(n)$, (55) gives

$$|e - m| \le \left|m - \frac{5}{32}N^2\right| + \frac{5}{2}v(N) \le \frac{5}{2}\Psi(m). \tag{56}$$

We also need the second-order cost of the integer balancing. For positive real class sizes x, define

$$Q(x) = c_* E(x)^{5/2} - P(x).$$

The Clebsch counts give $Q(\mathbf{1}) = 0$ and $\nabla Q(\mathbf{1}) = 0$. The function Q has a bounded Hessian in a fixed neighborhood of $\mathbf{1}$. Write $q = N/16$ and $n = q\mathbf{1} + h$, where $|h_i| \le 1$ and $\|h\|_2^2 = v(N)$. Homogeneity and Taylor's theorem yield, uniformly for large N,

$$0 \le Q(n) = q^5 Q(\mathbf{1} + h/q) \le Cq^3\|h\|_2^2 \le C'N^3 v(N). \tag{57}$$

The nonnegativity follows from the sharp weighted bound. If $v(N) = 0$, the same equation holds exactly, with both sides zero.

If $e \le m$, add a disjoint matching with $m - e$ edges. If $e > m$, delete any $e - m$ edges of H. A single edge deletion can destroy at most $\binom{N-2}{3}$ induced

pentagons, since every affected pentagon must contain both endpoints of that edge. Thus the resulting graph G_m has exactly m edges and

$$I_5(G_m) \geq P(n) - |e-m|\binom{N-2}{3}.$$

Since $e = m + O(\sqrt{m})$ and $N = \Theta(\sqrt{m})$, the mean value theorem, (56), and (57) imply

$$\begin{aligned}\Delta(m) &\leq c_* m^{5/2} - I_5(G_m) \\ &\leq c_* |m^{5/2} - e^{5/2}| + Q(n) + |e-m|\binom{N-2}{3} \\ &\leq C'' m^{3/2}\big(|e-m| + v(N)\big) \leq b_0 m^{3/2}\Psi(m).\end{aligned}$$

This proves (51).

Finally, $\Psi(m) = 0$ requires $v(N) = 0$, hence $16 \mid N$, and then $m = 5N^2/32 = 40(N/16)^2$. The converse is immediate. If m is not of this form, every term in the minimum is at least $15/16$: for $16 \mid N$ the nonzero integer $|m - 5N^2/32|$ is at least one, and otherwise $v(N) \geq 15/16$. Let $d = \operatorname{dist}(t, \mathbb{Z})$. For every $N \geq 0$,

$$\left| m - \frac{5}{32}N^2 \right| = \frac{5}{32}|t-N|(t+N) \geq \frac{5}{32}td.$$

Choosing an integer nearest to t gives

$$\Psi(m) \leq \frac{5}{32}(2t + 1/2)d + 4.$$

Together with the lower bound $15/16$, these inequalities show $\Psi(m) \asymp \sqrt{m}\, d + 1$ outside the equality sequence. Equation (52) follows. □

Corollary 10.2 (The m^2 error is uniformly optimal)**.** *For $m_k = 40k^2 + \lfloor 5k/2 \rfloor$,*

$$\Delta(m_k) = \Theta(m_k^2).$$

For each fixed nonzero integer j and sufficiently large k,

$$\Delta(40k^2 + j) = \Theta(|j|(40k^2 + j)^{3/2}).$$

At $m = 40k^2$ the deficit is exactly zero.

Proof. For the first sequence, the minimizing index in Ψ is $N = 16k$, and $\Psi(m_k) = \lfloor 5k/2 \rfloor$. Indeed, the next larger square value is $40k^2 + 5k + 5/32$, and all smaller indices have a strictly greater distance from m_k than $N = 16k$; the nonnegative term $v(N)$ cannot reverse this. For fixed $j \neq 0$, the same index minimizes for all sufficiently large k, with $\Psi(40k^2 + j) = |j|$. Apply Theorem 10.1. The last assertion is the exact Clebsch construction. □

Remark 10.3. Theorem 10.1 determines the order of the finite-edge deficit, with constants uniform in m. It does not identify the exact maximizing graph at each integer edge count. Theorem 12.2 below identifies the leading

coefficient on the m^2 scale. Its uniform absolute remainder does not replace the present comparison at phases extremely close to zero.

11. Root profiles and the two arithmetic constructions

Write $P(x)$ for the Clebsch pentagon polynomial and A for its adjacency matrix. An added vertex has fractional adjacency proportions $p = (p_i)_{i=1}^{16} \in [0,1]^{16}$ to sixteen unit classes. Its degree and pentagon link are

$$d(p) = \sum_i p_i, \qquad c(p) = \sum_{\mathrm{Cl}[U] \cong P_4} \prod_{\substack{i \text{ endpoint} \\ \text{of } U}} p_i \prod_{\substack{j \text{ internal} \\ \text{in } U}} (1 - p_j).$$

Lemma 11.1 (The profile envelope)**.** *For every fractional profile,*

$$c(p) \le \min\left\{ \frac{12}{5} d(p)^2, 60 \right\}.$$

For every $0 \le d \le 5$, the first bound is attained by setting $p_i = d/5$ on a Clebsch neighborhood and zero elsewhere.

Proof. Every nonadjacent pair of Clebsch vertices is the endpoint pair of exactly six induced four-vertex paths; adjacent pairs are endpoints of none. Dropping the internal nonneighbor factors therefore gives

$$c(p) \le 6 \sum_{\substack{i<j \\ ij \notin E(\mathrm{Cl})}} p_i p_j.$$

The independence number of Cl is five. Indeed, after translating an independent set to contain 0, its other vertices lie in the ten pair vertices of the standard labeling. Pairwise nonadjacency there means pairwise intersection of two-subsets of [5]. Such a family has at most four members, since a family of size four must be a star. Clebsch neighborhoods attain size five.

For completeness, the weighted clique inequality of Motzkin and Straus [10] used here follows by repeatedly merging the weights of two nonadjacent vertices in the support of a graph. With their sum fixed, the edge quadratic form is linear in the two weights, so one of the two mergers does not decrease it. Eventually the support is a clique of size r, where its value is at most $(1 - 1/r)d^2/2$ by Cauchy–Schwarz. Apply this to the complement of Cl, whose clique number is five, to obtain the first bound.

The bound $c(p) \le 60$ is a finite exact ingredient: for every one of the $65,536$ Boolean neighborhoods N, count the 480 induced P_4 subsets with endpoints in N and internal vertices outside N. The supplied standard-library program `verify_profile_envelope.py` constructs these paths from the Cayley edge list and checks every count. It finds $c(N) \le 60$, with equality precisely at the sixteen open and sixteen closed neighborhoods. The degreewise maxima, for degrees zero through sixteen, are

$$0, 0, 6, 18, 36, 60, 60, 54, 48, 44, 42, 35, 25, 18, 12, 0, 0.$$

Multilinearity represents $c(p)$ as the expectation of $c(N)$ for an independent Bernoulli subset, so the Boolean inequality gives the fractional one.

Finally, a neighborhood is an independent five-set. An induced four-vertex path with endpoints in that neighborhood has its two internal vertices outside it. There are six paths per endpoint pair and ten pairs, yielding $c(p) = 60(d/5)^2$ for the stated profile. □

Define

$$x_m = \sqrt{\frac{32m}{5}}, \qquad \alpha_m = x_m - \lfloor x_m \rfloor, \qquad k_m = \sqrt{\frac{m}{40}} = \frac{x_m}{16},$$

and put

$$f(\alpha) = \frac{(1-\alpha)\min\{3\alpha, 1\}}{80}, \qquad 0 \le \alpha < 1.$$

As usual $\Delta(m) = c_* m^{5/2} - \rho(C_5, m)$.

Proposition 11.2 (Uniform constructive upper error)**.** *As $m \to \infty$ through the integers,*

$$\Delta(m) \le m^2 f(\alpha_m) + O(m^{7/4}),$$

with an absolute implicit constant.

Proof. Write $k = k_m$ and $\alpha = \alpha_m$. All estimates below are uniform for $0 \le \alpha < 1$.

For the first construction let $N = \lfloor x_m \rfloor$ and distribute N vertices into sixteen classes whose sizes differ by at most one. Their Clebsch blow-up B has

$$e(B) = \frac{5N^2}{32} + O(1), \qquad I_5(B) = 192\left(\frac{N}{16}\right)^5 + O(k^3).$$

Fix a template vertex v. Add a single new vertex, with neighbors only in the five classes indexed by $N_{\mathrm{Cl}}(v)$, and choose its degree as close to $m - e(B)$ as feasibility permits. This leaves at most $O(1)$ edges to add or delete: the available number of neighbors is $5k + O(1)$, whereas $m - e(B) = 5k\alpha + O(1)$. Within those five classes choose the neighbor counts to be $\alpha k + O(1)$, with their total equal to the chosen degree. Such simultaneous rounding is possible by distributing the bounded integer remainder among the five classes.

There are sixty induced path templates with their endpoints in the selected neighborhood. Consequently the added vertex lies in $60\alpha^2 k^4 + O(k^3)$ induced pentagons. Correct the bounded edge remainder by adding a disjoint matching or deleting edges. Each deletion destroys at most $O(k^3)$ of the existing pentagons; adding the matching changes none. The resulting graph has exactly m edges and

$$\begin{aligned} I_5(G) &= 192(k - \alpha/16)^5 + 60\alpha^2 k^4 + O(k^3) \\ &= 192k^5 - 60\alpha(1-\alpha)k^4 + O(k^3). \end{aligned}$$

Thus $\Delta(m) \le 60\alpha(1-\alpha)k^4 + O(k^3)$.

For the second construction put $N = \lfloor x_m \rfloor + 1$, $\beta = 1 - \alpha$, and $t = N/16 = k + \beta/16$. The Clebsch matrix has a zero-sum vector $w \in \{-1, 1\}^{16}$ satisfying

$Aw = -3w$. Explicitly, at the vertex $u \in \mathbb{F}_2^4$ take $w_u = (-1)^{u_1+u_2+u_3}$; it has squared norm sixteen. Set

$$r^2 = \frac{40t^2 - m}{24}, \qquad y_i = t + rw_i.$$

Then $\sum_i y_i = N$, all y_i are positive for large m, and

$$E(y) = 40t^2 - 24r^2 = m, \qquad \sum_i (y_i - t)^2 = 16r^2 = \frac{2}{3}(40t^2 - m).$$

Round the y_i to integers n_i preserving their sum N and with $|n_i - y_i| \le 1$. Since $r = O(\sqrt{k})$ and the rounding error has sum zero, the resulting blow-up B satisfies

$$e(B) = m + O(\sqrt{k}), \qquad V := \sum_i (n_i - t)^2 = \frac{2}{3}(40t^2 - m) + O(\sqrt{k}).$$

The logarithmic Hessian calculation for the template, or direct polynomial expansion, gives uniformly for zero-sum $h = O(\sqrt{k})$,

$$c_* E(t\mathbf{1} + h)^{5/2} - P(t\mathbf{1} + h) = 6t^3 \|h\|_2^2 + O(k^{7/2}).$$

Hence the deficit of B at its own edge count is

$$6k^3 \frac{2}{3}(40t^2 - m) + O(k^{7/2}) = 20\beta k^4 + O(k^{7/2}).$$

Correct $e(B) - m$ by adding a disjoint matching or deleting edges. The number of changes is $O(\sqrt{k})$, and each deletion destroys at most $O(k^3)$ pentagons. The change in the comparison expression $c_* e(B)^{5/2}$ is also $O(k^{7/2})$. Thus this construction gives $\Delta(m) \le 20(1-\alpha)k^4 + O(k^{7/2})$.

Take the better construction and use $k^4 = m^2/1600$. □

12. A sharp arithmetic phase law

We retain $k = \sqrt{m/40}$, $x = 16k = \sqrt{32m/5}$, $\alpha = x - \lfloor x \rfloor$, and the function

$$f(\alpha) = \frac{(1-\alpha)\min\{3\alpha, 1\}}{80}$$

from the preceding section.

Lemma 12.1 (The scalar profile problem)**.** *Put $\phi(u) = \max\{12u, -4u\}$. For arbitrary fractional profiles $p_1, \dots, p_b \in [0,1]^{16}$ and any integer ℓ,*

$$\sum_{j=1}^{b} (12d(p_j) - c(p_j)) + \phi\left(5(\alpha + \ell) - \sum_{j=1}^{b} d(p_j)\right) \ge 20(1-\alpha)\min\{3\alpha, 1\}. \tag{58}$$

Proof. Write $z_j = d(p_j)/5$ and $\Phi(u) = \phi(5u)$. Lemma 11.1 gives

$$12d(p_j) - c(p_j) \ge g(z_j), \qquad g(z) = \begin{cases} 60z(1-z), & 0 \le z \le 1, \\ 60(z-1), & z \ge 1. \end{cases}$$

The function Φ is Lipschitz with constant sixty. Thus replacing any $z_j > 1$ by one can only decrease $\sum_j g(z_j) + \Phi(\alpha + \ell - \sum_j z_j)$: its profile cost drops

by $60(z_j - 1)$ and the other term rises by at most that amount. We may consequently suppose $0 \le z_j \le 1$.

For a fixed sum $\sum_j z_j = a + t$, where a is an integer and $0 \le t < 1$, the sum $\sum_j z_j^2$ is maximized by a ones, one entry t, and zeros. One can see this by moving mass between any two entries strictly between zero and one until one reaches an endpoint. Hence

$$\sum_j 60 z_j(1 - z_j) \ge 60t(1 - t).$$

Absorbing a into ℓ reduces the problem to minimizing

$$60t(1 - t) + \Phi(\alpha + \ell - t), \qquad 0 \le t \le 1, \quad \ell \in \mathbb{Z}.$$

For $\ell \le -1$, the second term alone is at least $20(1 - \alpha)$; for $\ell \ge 1$, the minimum over t is at least 60α. For $\ell = 0$ and $t \le \alpha$ the expression is $60(\alpha - t^2)$, whose minimum is $60\alpha(1 - \alpha)$. For $t \ge \alpha$ it is the concave quadratic $80t - 60t^2 - 20\alpha$, so its minimum is attained at $t = \alpha$ or $t = 1$. The two values are $60\alpha(1 - \alpha)$ and $20(1 - \alpha)$, respectively. Their minimum is the asserted right side. □

Theorem 12.2 (Sharp phase-dependent second-order asymptotic)**.** *As $m \to \infty$ through the positive integers,*

$$\rho(C_5, m) = c_* m^{5/2} - \frac{m^2}{80}(1 - \alpha_m)\min\{3\alpha_m, 1\} + O(m^{15/8}),$$
$$\alpha_m = \left\{ \sqrt{\frac{32m}{5}} \right\}. \tag{59}$$

The implicit constant is absolute; in particular the remainder is uniform as the phase approaches either endpoint.

Proof. The constructive upper bound for the deficit is Proposition 11.2. We prove the matching lower bound with error $O(k^{15/4}) = O(m^{15/8})$. Choose an extremal graph G with m edges. The coarse lower construction gives $\Delta(m) = O(m^2) = O(k^4)$, so the linear structural stability theorem supplies a Clebsch blow-up H, with isolates, on the same vertex set such that

$$T := |E(G) \triangle E(H)| = O(k), \qquad N_H = 16k + O(1), \qquad n_i^H = k + O(\sqrt{k}), \tag{60}$$

and $\sum_i (n_i^H - N_H/16)^2 = O(k)$. All constants in this proof are independent of m and α.

Discard vertices isolated in both G and H. Every remaining isolated vertex of H is incident with an edge in $E(G) \setminus E(H)$, so there are at most $2T$ such vertices. The total number of remaining vertices is therefore $O(k)$. This observation allows the counting estimates below to be independent of the original number of isolates.

Let the wrong degree of a vertex be its number of incident pairs in $E(G) \triangle E(H)$. Call a vertex bad if its wrong degree exceeds $k^{3/4}$. The bad

set B has size

$$b \le 2T/k^{3/4} = O(k^{1/4}).$$

Let G_0, H_0 be the graphs remaining after deleting B. Write n_i for the sixteen active class sizes of H_0, put $N = \sum_i n_i$, $q = N/16$, and $h_i = n_i - q$. Then

$$q = k + O(k^{1/4}), \qquad n_i = k + O(\sqrt{k}), \qquad V := \sum_i h_i^2 = O(k). \tag{61}$$

In particular every class is nonempty for all sufficiently large m. If A is the Clebsch adjacency matrix, put

$$a = \frac{h^T A h}{2k}, \qquad v = V/k.$$

The identity $A^2 = 3I - 2A + 2J$ shows that the eigenvalues on the sum-zero subspace are 1 and -3. Consequently

$$-\frac{3}{2}v \le a \le \frac{1}{2}v. \tag{62}$$

For a bad vertex j, let p_{ji} be the proportion of the ith active class of H_0 adjacent to j in G. Denote this profile by p_j, and abbreviate $d_j = d(p_j)$, $c_j = c(p_j)$. Let t_- count the edges of H_0 missing from G_0, and let t_+ count the edges added to H_0 inside G_0, together with the edges from B to the isolated class of H_0. Both are $O(k)$: the latter additional edges are all wrong pairs in the original comparison with H. Let t_n count those added edges inside G_0 whose endpoints have distinct nonadjacent active types; thus $0 \le t_n \le t_+$.

The edge count is

$$\frac{m - 5N^2/32}{k} = a + \sum_{j \in B} d_j + \frac{t_+ - t_-}{k} + O(k^{-1/4}). \tag{63}$$

Indeed, $e(H_0) = 5N^2/32 + ka$. The exact number of edges from a bad vertex to the active classes is $\sum_i n_i p_{ji}$; replacing this by kd_j has error $O(\sqrt{k})$ per bad vertex and thus $O(k^{3/4})$ in total. There are at most $b^2 = O(k^{1/2})$ edges within B. These are all the remaining terms.

We next expand the pentagon count and give the needed error estimates explicitly. Every good vertex has wrong degree at most $k^{3/4}$. If r_u denotes its wrong degree in the comparison of G_0 and H_0, then

$$\sum_u r_u^2 \le k^{3/4} \sum_u r_u = O(k^{7/4}).$$

An intersecting pair of wrong edges fixes three vertices and belongs to $O(k^2)$ five-sets; a disjoint pair fixes four vertices and belongs to $O(k)$ five-sets. The number of five-sets containing at least two wrong edges is therefore

$$O\left(k^2 \sum_u r_u^2 + kT^2\right) = O(k^{15/4}). \tag{64}$$

For a five-set with at most one wrong pair, the change of its induced-pentagon indicator is exactly its one-edge change. For every other five-set, the difference between the actual change and the sum of its one-edge changes has absolute value at most eleven. Hence (64) bounds the entire higher-order error.

By the two-vertex coefficients of Lemma A.3, one deleted template edge loses $24k^3 + O(k^{5/2})$ pentagons, and an added edge between distinct nonadjacent active types has change $-6k^3 + O(k^{5/2})$. Edges added within a class or meeting an isolated type have one-edge change zero. The estimates are uniform because of (61); excluding the two fixed endpoints from the available old classes introduces only $O(k^2)$ further error. There are $O(k)$ wrong edges. Thus

$$I_5(G_0) = P(n) - 24k^3 t_- - 6k^3 t_n + O(k^{15/4}). \tag{65}$$

Pentagons containing at least two bad vertices number at most $O(b^2k^3) = O(k^{7/2})$. For a fixed bad vertex, changing the background on the other four vertices from G_0 to H_0 affects at most $O(Tk^2)$ five-sets. Summing over B costs $O(bTk^2) = O(k^{13/4})$. In the perfect background H_0, the root link is obtained from $c(p_j)$ by giving its four distinct classes weights n_i. Each product of four such weights is $k^4 + O(k^{7/2})$. There are only 480 path templates, so this link is $k^4 c_j + O(k^{7/2})$, uniformly in the profile. Summing the latter errors over B gives $O(k^{15/4})$. Isolated background vertices lie in no such path. We conclude from (65) that

$$I_5(G) = P(n) + k^4 \sum_{j \in B} c_j - 24k^3 t_- - 6k^3 t_n + O(k^{15/4}). \tag{66}$$

It remains to expand the smooth template defect. For $Q(x) = c_* E(x)^{5/2} - P(x)$, the Clebsch counts yield

$$Q(\mathbf{1}) = 0, \qquad \nabla Q(\mathbf{1}) = 0, \qquad \nabla^2 Q(\mathbf{1}) = 12I - \tfrac{3}{4}J.$$

Homogeneity, $\sum_i h_i = 0$, and Taylor's theorem give

$$Q(n) = 6q^3 V + O(q^2 \|h\|_2^3) = 6k^3 V + O(k^{7/2}).$$

Also $e(H_0) = m + O(k^{5/4})$, by (61) and $V = O(k)$. Since $(5/2)c_* m^{3/2} = 12k^3$, another Taylor expansion gives

$$c_* m^{5/2} - c_* e(H_0)^{5/2} = 12k^3 \big(m - e(H_0)\big) + O(k^{7/2}).$$

Combining these identities with (63) and (66) produces

$$\frac{\Delta(m)}{k^4} = 6v + \sum_{j \in B} (12d_j - c_j) + 12\frac{t_+ + t_-}{k} + 6\frac{t_n}{k} + O(k^{-1/4}). \tag{67}$$

The scalar function ϕ is subadditive and Lipschitz with constant twelve. Equation (62) implies $6v \geq \phi(a)$, while

$$12(t_+ + t_-)/k \geq \phi\big((t_+ - t_-)/k\big).$$

Drop the nonnegative t_n term and apply (63). If $r = (m - 5N^2/32)/k$, we obtain

$$\frac{\Delta(m)}{k^4} \geq \sum_{j \in B} (12d_j - c_j) + \phi\left(r - \sum_{j \in B} d_j\right) - O(k^{-1/4}).$$

Since $N = 16k + O(k^{1/4})$,

$$\begin{aligned} r &= 5(16k - N) - \frac{5}{32k}(16k - N)^2 \\ &= 5(\alpha + \ell) + O(k^{-1/2}), \qquad \ell = \lfloor 16k \rfloor - N \in \mathbb{Z}. \end{aligned}$$

Apply Lemma 12.1 and use the Lipschitz bound once more. The result is

$$\Delta(m) \geq 20k^4(1 - \alpha)\min\{3\alpha, 1\} - O(k^{15/4}).$$

Because $m^2 = 1600k^4$, this is the desired lower bound. Proposition 11.2 has an upper error $O(m^{7/4})$, which is absorbed in $O(m^{15/8})$. □

Corollary 12.3 (The optimal uniform second-order coefficient)**.** *The finite-edge deficit satisfies*

$$\limsup_{m \to \infty} \frac{\Delta(m)}{m^2} = \frac{1}{120}, \qquad \liminf_{m \to \infty} \frac{\Delta(m)}{m^2} = 0.$$

Its set of subsequential limits is the whole interval $[0, 1/120]$*. In particular, the coefficient* $1/120$ *in the uniform estimate*

$$\rho(C_5, m) \geq c_* m^{5/2} - \frac{m^2}{120} - O(m^{15/8})$$

cannot be decreased.

Proof. The function f is continuous on the unit circle, vanishes at phase zero, and has maximum $1/120$ at phase $1/3$. For each fixed $0 < \beta < 1$, choose an integer m_k nearest to $(5/32)(16k + \beta)^2$. Factoring the difference of squares gives $\sqrt{32m_k/5} = 16k + \beta + O(k^{-1})$, so $\alpha_{m_k} \to \beta$. The exact equality sequence $m = 40k^2$ provides phase zero. The theorem therefore yields all values of the continuous range $f([0, 1]) = [0, 1/120]$ and excludes all other limit points. □

13. The lattice correction at a fixed edge offset

The phase law describes the m^2 scale. In a bounded edge neighborhood of the equality sequence, the deficit instead has order $m^{3/2}$. Its leading coefficient is a finite lattice optimization determined by the Clebsch adjacency matrix.

For $h \in \mathbb{Z}^{16}$ with $\sum_i h_i = 0$, put

$$Q(h) = \sum_{uv \in E(\mathrm{Cl})} h_u h_v = \tfrac{1}{2} h^T A h.$$

For an integer j, define

$$\Lambda(j) = \min_{\substack{h\in\mathbb{Z}^{16}\\ \sum_i h_i=0}} \left(6\|h\|_2^2 + 12|j - Q(h)|\right). \tag{68}$$

The minimum may be restricted to $\|h\|_2^2 \le 2|j|$: the zero vector gives value $12|j|$, and every vector outside that ball has strictly greater cost. In particular this is a finite, explicitly decidable optimization for each j.

Theorem 13.1 (Fixed-offset asymptotic)**.** *For every fixed integer j, as the positive integer k tends to infinity,*

$$\rho(C_5, 40k^2 + j) = c_*(40k^2 + j)^{5/2} - \Lambda(j)k^3 + O_j(k^2). \tag{69}$$

The coefficients satisfy

$$\Lambda(j) = 12j \quad (j \ge 0), \qquad 4|j| \le \Lambda(j) \le 12|j| \quad (j < 0),$$

and $\Lambda(-12) = 48$. Thus the fixed-offset correction is not symmetric under $j \mapsto -j$.

Proof. Write $m = 40k^2 + j$ and $\Delta(m) = c_* m^{5/2} - \rho(C_5, m)$. Start from the balanced blow-up of class size k. If $j \ge 0$, add a disjoint matching with j edges; if $j < 0$, delete $|j|$ edges. Since a single deletion affects $O(k^3)$ pentagons, this gives the preliminary estimate $\Delta(m) = O_j(k^3)$.

Let G be extremal with m edges. Its normalized deficit is $O_j(k^{-2})$, so Theorem 9.3 supplies a Clebsch blow-up H, with isolates, for which

$$|E(G) \triangle E(H)| = O_j(1), \qquad \sum_i (n_i - N/16)^2 = O_j(1),$$

$$N = \sqrt{32m/5} + O_j(k^{-1}) = 16k + O_j(k^{-1}).$$

The integer N therefore equals $16k$ for all sufficiently large k. Consequently $n_i = k + h_i$ for an integer vector h satisfying $\sum_i h_i = 0$ and $\|h\|_2 = O_j(1)$, and

$$e(H) = 40k^2 + Q(h). \tag{70}$$

After discarding common isolates, the two graphs have $O(k)$ vertices: every remaining isolated vertex of H is incident with one of the $O_j(1)$ wrong edges.

Let t_+ and t_- be the numbers of added and deleted edges in passing from H to G, and let t_n count added edges between distinct nonadjacent active classes. All these numbers are $O_j(1)$, and

$$j - Q(h) = t_+ - t_-.$$

The one-edge pentagon changes are $-24k^3 + O_j(k^2)$ for a deleted template edge, $-6k^3 + O_j(k^2)$ for an added edge of the latter nonadjacent type, and zero for an edge within a class or meeting an isolated type. Five-sets containing at least two wrong pairs number $O_j(k^2)$, by the same pair counting used in (64). Therefore

$$I_5(G) = P(k\mathbf{1} + h) - 24k^3 t_- - 6k^3 t_n + O_j(k^2).$$

The template Hessian gives

$$c_* e(H)^{5/2} - P(k\mathbf{1} + h) = 6k^3\|h\|_2^2 + O_j(k^2).$$

Since $m - e(H) = j - Q(h) = O_j(1)$, the comparison between the two edge powers is

$$c_* m^{5/2} - c_* e(H)^{5/2} = 12k^3(j - Q(h)) + O_j(k).$$

Combining these formulas yields

$$\begin{aligned}\Delta(m) &= k^3(6\|h\|_2^2 + 12(t_+ + t_-) + 6t_n) + O_j(k^2)\\ &\geq k^3(6\|h\|_2^2 + 12|j - Q(h)|) - O_j(k^2)\\ &\geq \Lambda(j)k^3 - O_j(k^2).\end{aligned}$$

For the reverse inequality, fix a minimizing vector h in (68). The class sizes $k + h_i$ are positive for large k. If $j - Q(h) > 0$, add that many edges within a class; if it is negative, delete its absolute value in template edges. There is ample capacity for either operation once k is large. All added edges are within a class, so $t_n = 0$, and the number of edits is exactly $|j - Q(h)|$. The same expansion gives a graph with exactly m edges whose deficit is $\Lambda(j)k^3 + O_j(k^2)$. This proves (69).

On the sum-zero subspace, the eigenvalues of A are 1 and -3. Hence

$$-\tfrac{3}{2}\|h\|_2^2 \leq Q(h) \leq \tfrac{1}{2}\|h\|_2^2.$$

For $j \geq 0$ this implies

$$6\|h\|_2^2 + 12|j - Q(h)| \geq 12Q(h) + 12|j - Q(h)| \geq 12j,$$

and $h = 0$ attains equality. For $j < 0$, it implies

$$6\|h\|_2^2 + 12|j - Q(h)| \geq -4Q(h) + 4|j - Q(h)| \geq 4|j|.$$

The upper bound $12|j|$ again follows from $h = 0$.

To evaluate the example $j = -12$, use the Cayley labeling $u \in \mathbb{F}_2^4$ and the characters $\chi_a(u) = (-1)^{a\cdot u}$. For the bit masks $a = 7$ and $a = 11$, each character has Clebsch eigenvalue -3. The vector

$$h = \tfrac{1}{2}(\chi_7 + \chi_{11})$$

is integral, has sum zero, has eight nonzero entries in $\{-1, 1\}$, and satisfies $Ah = -3h$. Thus $\|h\|_2^2 = 8$, $Q(h) = -12$, and its objective value is 48. The lower bound $4|j|$ proves that this value is optimal. □

Corollary 13.2 (Two exact finite-edge values)**.** *For all sufficiently large integers k,*

$$\rho(C_5, 40k^2 + 1) = 192k^5, \qquad \rho(C_5, 40k^2 - 1) = 192k^5 - 24k^3.$$

Up to isolated vertices, all extremal graphs at $40k^2 + 1$ edges are obtained from the balanced Clebsch blow-up of class size k by adding one edge either within a class or incident with an isolated vertex. At $40k^2 - 1$ edges, all extremal graphs are either a balanced blow-up with one edge deleted, or a

Clebsch blow-up in which two adjacent classes have sizes $k+1, k-1$ and the other fourteen classes have size k.

Proof. For $j = 1$, the lattice minimum is 12 by the theorem. For $j = -1$, it is also 12: the zero vector attains this value, and every nonzero integral sum-zero vector has squared norm at least two, so its first cost term alone is at least twelve.

For an extremal graph and its template from the proof above, write

$$L = 6V + 12(t_+ + t_-) + 6t_n.$$

This is an integer multiple of six and is at least twelve. The same proof gives $\Delta(m) = Lk^3 + O(k^2)$, with a constant uniform over the bounded family of permitted h and wrong-edge counts. The elementary constructions for $j = \pm 1$ have deficit $12k^3 + O(k)$. Hence $L > 12$, which would imply $L \geq 18$, is impossible for all sufficiently large k.

The nonnegative integer solutions to $L = 12$ are therefore $V = 0$, $t_+ + t_- = 1$, $t_n = 0$, or $V = 2$, $t_+ + t_- = t_n = 0$. Here we used that an integer sum-zero vector has even squared norm. In the second case h has one entry 1, one entry -1, and all others zero, so $Q(h)$ is zero or -1.

For $j = 1$, only the first case is possible. The single modification is an added edge, and $t_n = 0$ says that it lies within a class or meets an isolated vertex. Such an edge changes no pentagon count: in the first case its endpoints are true twins, and in the other cases any formerly isolated endpoint has degree one. This gives the asserted value and classification.

For $j = -1$, the first case deletes one edge of the balanced template. Exactly $24k^3$ old pentagons use that edge. There are no newly induced pentagons: the deleted endpoints have no common neighbor, whereas a nonadjacent pair in a five-cycle must have one. In the second case $Q(h) = -1$ forces the two perturbed classes to be adjacent. Multilinearity of the pentagon polynomial gives exactly

$$P(k\mathbf{1} + h) = 192k^5 + 60k^4 \sum_i h_i - 24k^3 = 192k^5 - 24k^3.$$

These constructions attain the common value, completing both the upper bound and the equality classification. □

14. Eventual polynomiality at every fixed offset

The lattice coefficient has a further consequence. For each fixed edge offset, the exact extremal function is eventually a polynomial, and the polynomial can be determined by a finite enumeration. This statement does not provide an explicit threshold beyond which the polynomial is valid.

Theorem 14.1 (Computable eventual polynomial)**.** *For every integer j, there are computable integers a_j, b_j, c_j and an integer $k_0(j)$ such that, for every integer $k \geq k_0(j)$,*

$$\rho(C_5, 40k^2 + j) = 192k^5 + \big(12j - \Lambda(j)\big)k^3 + a_j k^2 + b_j k + c_j. \tag{71}$$

Moreover, a finite procedure produces a finite list of graph constructions that, for all sufficiently large k, contains all extremal graphs up to isomorphism and the addition of isolates. The proof determines the eventual polynomial and this list; it does not give an explicit value or an algorithm for $k_0(j)$.

Proof. We first strengthen the discrete conclusion in the fixed-offset argument. Its template has integer class sizes $k + h_i$ with $\sum_i h_i = 0$ and bounded $V = \|h\|_2^2$, and a bounded number $T = t_+ + t_-$ of wrong pairs. As in that proof, put

$$L = 6V + 12T + 6t_n.$$

The edge-offset constraint and the definition of $\Lambda(j)$ give $L \geq \Lambda(j)$. Both numbers are integer multiples of six. The same uniformly bounded-template expansion gives

$$\Delta(40k^2 + j) = Lk^3 + O_j(k^2).$$

On the other hand, the construction in Theorem 13.1 gives $\Delta(40k^2 + j) \leq \Lambda(j)k^3 + O_j(k^2)$. The positive gap between distinct multiples of six therefore forces

$$L = \Lambda(j) \tag{72}$$

for every extremal graph and all sufficiently large k. Since $\Lambda(j) \leq 12|j|$, this implies the explicit bounds

$$V \leq 2|j|, \qquad T \leq |j|. \tag{73}$$

The initial boundedness and the uniform error in this argument come from strong stability, before imposing the explicit bounds (73); thus there is no circular assumption about the extremal family.

We now describe a finite list depending only on j. Enumerate:

(1) integer vectors $h \in \mathbb{Z}^{16}$ with $\sum_i h_i = 0$ and $\|h\|_2^2 \leq 2|j|$;
(2) a set X of $s \leq 2|j|$ labeled exceptional vertices, each assigned a type $\tau(x) \in \{0, 1, \ldots, 16\}$;
(3) a set $F \subseteq \binom{X}{2}$ of at most $|j|$ wrong pairs, with every vertex of X incident with a member of F.

Type zero is the isolated type. The case $F = X = \varnothing$ is allowed. The template adjacency between two exceptional vertices is Clebsch adjacency between their positive types, and is zero if either type is zero. A wrong pair toggles this template adjacency. Thus F determines t_+, t_-, and t_n without any further choices. Retain precisely the data satisfying

$$Q(h) + t_+ - t_- = j, \qquad 6\|h\|_2^2 + 12|F| + 6t_n = \Lambda(j). \tag{74}$$

All these tests use integer arithmetic, and the enumeration is finite with explicit bounds.

Each retained datum defines a graph for all sufficiently large k. Let e_i be the number of exceptional vertices of type i, and put $k + h_i - e_i$ ordinary vertices in each active class i. Together with the exceptional vertices, the active class sizes of the underlying template are exactly $k + h_i$. Include the

e_0 exceptional vertices of type zero, place all template edges, and toggle the pairs in F. The counts $k + h_i - e_i$ are nonnegative for sufficiently large k; this capacity requirement has an explicit threshold for each datum. The edge count is exactly $40k^2 + j$ by (74). Further isolates may be added freely.

Conversely, every extremal graph for sufficiently large k has such a representation. Take the endpoints of all its wrong pairs relative to the strong-stability template as X. There are at most $2T \le 2|j|$ of them. Outside X, all adjacencies agree with the template, and the remaining type-zero vertices are isolated. The bounds and both admissibility tests follow from (73), the exact edge count, and (72).

For each retained datum D, its pentagon count is an integer coefficient polynomial $P_D(k)$. To see this without an automorphism division, take a subset $S \subseteq X$ of exceptional vertices and a subset $I \subseteq [16]$ of ordinary types, with $|S| + |I| = 5$. Use one formal ordinary representative of every type in I, and test whether the resulting five-vertex graph is an induced pentagon. Let its indicator be $b_D(S, I)$. All ordinary vertices of the same type are false twins, even in their adjacencies to X, since wrong pairs have both endpoints in X. An induced pentagon cannot contain two such vertices. It follows that the exact count is

$$P_D(k) = \sum_{\substack{S \subseteq X,\ I \subseteq [16] \\ |S|+|I|=5}} b_D(S, I) \prod_{i \in I} (k + h_i - e_i). \tag{75}$$

Each five-set is counted once. This formula establishes both integrality of the coefficients and an explicit procedure to compute them.

Every retained datum has the same three leading coefficients. Indeed, the fixed-offset expansion applies to each of these fixed finite modifications and gives

$$c_*(40k^2 + j)^{5/2} - P_D(k) = \Lambda(j)k^3 + O_j(k^2).$$

Since

$$c_*(40k^2 + j)^{5/2} = 192k^5 + 12jk^3 + O_j(k),$$

the polynomial P_D has coefficient 192 at degree five, zero at degree four, and $12j - \Lambda(j)$ at degree three. Write its remaining coefficients as (a_D, b_D, c_D).

The retained family is nonempty. A minimizing lattice vector in (68), followed by the addition of within-class edges or deletion of template edges as in the fixed-offset proof, supplies such a datum. The wrong pairs may be chosen disjoint; their number is at most $|j|$, and their endpoints satisfy the stated bound.

For all sufficiently large k, every retained graph is feasible, and every extremal graph is represented. Thus

$$\rho(C_5, 40k^2 + j) = \max_D P_D(k).$$

Among finitely many polynomials, the one with lexicographically largest coefficient vector, read from highest degree to lowest, is maximal for all sufficiently large arguments. Choose the largest triple (a_D, b_D, c_D) and denote

it by (a_j, b_j, c_j). This proves (71). Retaining all data whose polynomial equals that maximum gives exactly the eventual extremal constructions, allowing duplicate isomorphic presentations. For $j = 0$ the list consists just of the balanced template, giving $192k^5$ as required.

Finally, the finite polynomial comparisons and the capacity thresholds can be computed from the list. The proof that all extremal graphs enter that list uses the qualitative compactness step in strong stability. We have not quantified its threshold, so the present procedure computes the eventual polynomial and the eventual construction list, rather than certifying their validity at an arbitrary supplied finite value of k. □

14.1. **Explicit eventual polynomials at small offsets.** The finite procedure can be carried out directly for the first five offsets. Its results, valid for all sufficiently large integers k, are

j	$\Lambda(j)$	$\rho(C_5, 40k^2 + j)$
-2	24	$192k^5 - 48k^3 + 6k^2$
-1	12	$192k^5 - 24k^3$
0	0	$192k^5$
1	12	$192k^5$
2	24	$192k^5 + 6k^2$

The cases $j = 0, \pm 1$ were proved above. We explain both the finite check and matching constructions for $j = \pm 2$.

For both signs, the lattice energy is 24. For $j = 2$ this follows from Theorem 13.1. For $j = -2$, a nonzero sum-zero integer vector has $V \geq 2$; if $V = 2$, its edge quadratic form is zero or -1, and its total lattice cost is at least 24. Vectors with $V \geq 4$ have first cost term at least 24, while the zero vector attains 24.

At an extremum the energy equality therefore leaves only

$$(V, T) = (0, 2), (2, 1), (4, 0), \qquad t_n = 0.$$

For $V = 2$ the class deviation vector has one entry 1 and one entry -1. For $V = 4$ it has two entries of each sign. Substituting these possibilities in the exact polynomial formula (75) gives only the corrections

$$\{0, 6k^2\}$$

relative to $192k^5$ when $j = 2$. When $j = -2$, the corrections relative to $192k^5 - 48k^3$ are precisely

$$\{0, 2k, 6k^2, -6k^2, -2k^2 + 2k, 2k^2 + 2k, -2k^2, 2k^2\}.$$

The enumeration uses the following convention. The sixteen Clebsch types remain labeled, and the deviation vector h is not identified under Clebsch automorphisms. Exceptional vertices are identified only under isomorphisms of their wrong-pair graph F that preserve their types. Thus a single wrong pair has unordered endpoint types, a P_3 has a fixed center and unordered leaf types, and $2K_2$ has two unordered endpoint pairs that are themselves

unordered. Repeated types are allowed; the exceptional vertices themselves are distinct. This convention gives

(V,T)	$j=2$	$j=-2$
$(0,2)$	762	$1,060$
$(2,1)$	0	$3,200$
$(4,0)$	120	$1,200$
Total	882	$5,460$

Here is a direct derivation of every entry. An allowed added edge has both endpoints in one active class or at least one endpoint of isolated type. There are $16+17=33$ unordered type pairs of this kind. With $V=0$ and $j=2$, two disjoint additions therefore give $\binom{34}{2}=561$ presentations. For incident additions, an isolated center gives $\binom{18}{2}=153$ unordered leaf-type pairs. An active center has only its own type and the isolated type available for either leaf, giving $16\binom{3}{2}=48$ presentations. The first total is consequently $561+153+48=762$. For $j=-2$, a deleted edge has one of the 40 Clebsch edge types. Two disjoint deletions give $\binom{41}{2}=820$ presentations, while incident deletions give $16\binom{6}{2}=240$. Their sum is $1,060$.

For $V=2$, write $h=e_u-e_v$, so $Q(h)=-A_{uv}$ is zero or -1. One modification cannot then produce offset 2. To produce offset -2, the ordered pair (u,v) must be adjacent and the modification must be a deletion. There are $80\cdot 40=3,200$ choices.

For $V=4$, the support of h consists of two positive and two negative vertices. The Clebsch graph has 60 induced copies of $2K_2$ and 480 of P_3+K_1. To verify these counts directly, the six common nonneighbors of the endpoints of an edge induce a matching of size three: each has four neighbors in the union of the endpoint neighborhoods and hence one among those six vertices. Thus the first count is $40\cdot 3/2=60$. There are $16\binom{5}{2}=160$ induced paths of order three. The union of the neighborhoods of their three vertices has size 13, so each path has exactly three possible isolated fourth vertices. This gives $160\cdot 3=480$. The condition $Q(h)=2$ requires two edges within the sign classes and no cross edge, giving $2\cdot 60=120$ sign assignments. The condition $Q(h)=-2$ requires exactly two cross edges and no edge within a sign class: a within-class edge together with the required three or more cross edges would create a triangle. The two cross edges are either disjoint, with four balanced sign assignments, or form a path with an isolated vertex, with two. Hence the last negative-offset count is $4\cdot 60+2\cdot 480=1,200$.

The supplied integer-arithmetic programs independently check this branch census and evaluate (75) on every presentation. The two offsets give respectively two and eight distinct polynomials, exactly as listed above. For $j=-1,0,1$ the same convention gives $120,1,33$ presentations; there are $6,496$ in total over the five offsets. In both displayed correction lists for $j=\pm 2$, the maximum is $6k^2$ for every positive k. Eventual global extremality uses Theorem 14.1, not merely this finite comparison.

For an explicit construction at $j = 2$, add a new vertex to the balanced blow-up and join it to two old vertices in distinct nonadjacent Clebsch classes. There are six induced path templates between those types, and the two internal vertices can each be chosen in k ways. Thus exactly $6k^2$ induced pentagons are added. At $j = -2$, delete two edges with a common endpoint and their other endpoints in distinct classes. Each deleted edge lies in $24k^3$ pentagons, and exactly $6k^2$ pentagons contain both. Deletions create no new induced pentagons in the triangle-free template. This realizes $192k^5 - 48k^3 + 6k^2$ and completes the two examples.

Appendix A. Local stability in edit distance

We give a self-contained local argument. It permits fractional edge values and triangles, which may arise during the modifications in the proof. Accordingly, throughout this appendix put

$$P(W) = \frac{1}{10}\int_{[0,1]^5} \prod_{ij\in E(C_5)} W(x_i,x_j) \prod_{\substack{i<j\\ ij\notin E(C_5)}} (1-W(x_i,x_j))\,dx_1\cdots dx_5,$$

$$F(W) = \frac{P(W)}{E(W)^{5/2}}, \qquad R_{\mathrm{ind}}(W) = 5^{5/2}F(W).$$

These definitions agree with $P(W) = C(W)$ and $R_{\mathrm{ind}}(W) = R(W)$ for every graphon in the main text. Let W_0 be the Clebsch graphon on a fixed partition $A_1,\dots,A_{16}$ into sets of measure $a = 1/16$. As before, $\mathcal{E}_{\mathrm{Cl}}$ includes balanced Clebsch graphons on any positive active mass, with isolates on the complement. Set

$$d_1(W,\mathcal{E}_{\mathrm{Cl}}) = \inf_{U\in\mathcal{E}_{\mathrm{Cl}}} \|W-U\|_1.$$

All measurable partitions are allowed in the infimum.

Theorem A.1 (Local quadratic stability)**.** *There exist $\delta_0 > 0$ and $c_{\mathrm{loc}} > 0$ such that every graphon $W : [0,1]^2 \to [0,1]$ with $\|W - W_0\|_1 < \delta_0$ satisfies*

$$R_* - R_{\mathrm{ind}}(W) \geq c_{\mathrm{loc}}\, d_1(W,\mathcal{E}_{\mathrm{Cl}})^2. \tag{76}$$

Equality in $R_{\mathrm{ind}}(W) \leq R_*$ holds precisely when $W \in \mathcal{E}_{\mathrm{Cl}}$ up to null sets.

A.1. **Three finite properties of the Clebsch graph.** Write the five generators of Cl as $s_1,\dots,s_5$, with $\sum_i s_i = 0$. Their only nonempty dependence is the full five-element set. Relative to a vertex u, its neighbors are $u+s_i$ and its ten nonneighbors, other than itself, are $u+s_i+s_j$, indexed by $\{i,j\} \in \binom{[5]}{2}$. A pair vertex is adjacent to the two neighbors indexed by its elements; two pair vertices are adjacent exactly when their pairs are disjoint. In particular, the ten pair vertices induce the Petersen graph $\mathrm{KG}(5,2)$.

We will also use the following counts, including their normalizations. Every closed walk of length five uses each generator once, and is a simple pentagon. Hence Cl has $16\cdot 5!/10 = 192$ pentagons and each vertex belongs to $5!/2 = 60$ of them. A fixed edge belongs to $4! = 24$ pentagons: orient it from its

specified first endpoint and permute the remaining four generators. For a fixed nonedge from 0 to $s_i + s_j$, a rooted closed walk reaches the second endpoint at step two or at step three. There are 2!3! orders in each case. Dividing the total by the two orientations gives 12 pentagons through that pair.

Lemma A.2 (The neighborhood inequality)**.** *Add a vertex to* Cl *with neighborhood $N \subseteq V(\mathrm{Cl})$, and let $c(N)$ count the induced pentagons containing it. Then*

$$c(N) \le 12|N|.$$

Equality holds exactly when $N = \varnothing$ or $N = N_{\mathrm{Cl}}(v)$ for a vertex v.

Proof. Deleting the added vertex from a pentagon leaves an induced P_4 with its endpoints in N and internal vertices outside N. For $u \in N$, let $c_u(N)$ count these paths with endpoint u, so that $2c(N) = \sum_{u \in N} c_u(N)$.

Use the pair labeling relative to u. Let U be its ten pair vertices, $T = N \cap U$, and $D = \{i : u + s_i \in N\}$. A path counted by $c_u(N)$ has the form $u, u+s_i, b, v$, where $b \in U \setminus T$, $v \in T$, the pairs b, v are disjoint, and $i \in b \setminus D$. These conditions are also sufficient. Thus, writing $P = \mathrm{KG}(5,2)$,

$$c_u(N) = \sum_{b \in U \setminus T} (2 - |b \cap D|) d_T(b) \le 2e_P(T, U \setminus T). \tag{77}$$

Here the maximum cut of P has size twelve. To verify this directly, P has parameters $(10, 3, 0, 1)$: disjoint pairs have no common neighbor, while two intersecting pairs have the unique complementary pair as common neighbor. It has no four-cycle. For any fixed edge, choose one of the two other neighbors at each endpoint. The two choices are nonadjacent and have a unique common neighbor; they therefore complete the edge to a pentagon. This gives four pentagons through every edge, and $15 \cdot 4/5 = 12$ pentagons in total. Every cut leaves at least one edge of each odd cycle uncut. Since an edge lies in four pentagons, at least three of the fifteen edges are uncut. A star

$$F_i = \{\{i, j\} : j \ne i\}$$

is an independent four-set with twelve edges leaving it, so the bound is attained.

Moreover every twelve-edge cut has a side of size four, and that side is independent. Indeed, a side of size at most three has at most nine outgoing edges, and a side of size five has an odd number of outgoing edges. Four pairwise intersecting two-subsets form a star: if $\{1,2\}, \{1,3\}$ are present and some pair omits 1, that pair must be $\{2,3\}$ and there can be no fourth distinct pair. Thus the maximum cut sides are precisely F_i and $U \setminus F_i$.

Equation (77) proves $c_u(N) \le 24$. If equality holds and $T = F_i$, every pair outside this star has two neighbors in it. The loss term in (77) vanishes only if $D \subseteq \{i\}$. Consequently $N = N_{\mathrm{Cl}}(u + s_i)$ or $N = N_{\mathrm{Cl}}[u + s_i]$, of size five or six. If $T = U \setminus F_i$, every pair in F_i has three neighbors in T, and the pairs in

F_i use all five indices. Hence $D = \varnothing$ and $N = \{u\} \cup (U \setminus F_i)$ has size seven; in this case u has no neighbor in N.

Suppose $N \neq \varnothing$ and $c(N) = 12|N|$. Then every $u \in N$ has $c_u(N) = 24$. The size-six case is impossible: at the center of a closed neighborhood, the first internal vertex of a counted path cannot lie outside N, so its count is zero. The size-seven case is also impossible. The six pair vertices in that case contain an edge, since an independent set in $\mathrm{KG}(5, 2)$ has at most four vertices by the preceding intersecting-pair argument. An endpoint of this edge has a neighbor in N, whereas the equality classification at that endpoint permits a neighbor in N only in the size-six case. Therefore N is a vertex neighborhood. Conversely, adding a clone of a vertex creates exactly its sixty pentagons. No such pentagon can contain both clones, since C_5 has no pair of twins. Thus $c(N) = 60 = 12|N|$ for a vertex neighborhood, and the empty case is immediate. □

Lemma A.3 (Two-vertex coefficients)**.** *Add two vertices with old neighborhoods $N_{\mathrm{Cl}}(i)$, $N_{\mathrm{Cl}}(j)$, and prescribe whether they are adjacent. The numbers of induced pentagons containing both added vertices and three old vertices are*

	added pair absent	*added pair present*
$i = j$	0	0
$ij \in E(\mathrm{Cl})$	0	24
$i \neq j,\ ij \notin E(\mathrm{Cl})$	12	6

Proof. For equal types the added vertices are twins, whether adjacent or not, so cannot both occur in an induced pentagon. For distinct types, when their prescribed adjacency agrees with that of i, j, the counts are the ordinary pair counts 24 and 12 above. A pentagon cannot use two vertices from the same clone class, so the old vertices of types i, j introduce no extra completions. When ij is an edge but the added pair is absent, the two added vertices have no common old neighbor. This rules out a pentagon, in which every nonadjacent pair has a common neighbor.

For the last entry translate i to 0 and write $j = s_1 + s_2$. Their exclusive old neighbors are, respectively,

$$\{s_3, s_4, s_5\}, \qquad \{s_3 + s_4, s_3 + s_5, s_4 + s_5\}.$$

A pentagon containing the added edge must join one exclusive neighbor on either side through a common old vertex outside $N_{\mathrm{Cl}}(i) \cup N_{\mathrm{Cl}}(j)$. The two exclusive neighbors must be nonadjacent. Thus a pair from $\binom{\{3,4,5\}}{2}$ is matched with its unique missing generator, giving three choices. For example $s_3 + s_4$ and s_5 have common neighbors $s_1 + s_5, s_2 + s_5$, both outside the forbidden union. There are exactly two completions for each choice, giving six in total. □

Lemma A.4 (The template Hessian)**.** *For the Clebsch vertex-weight polynomials $P(x)$ and $E(x)$,*

$$\nabla^2 \log(P/E^{5/2})(\mathbf{1}) = \frac{J - 16I}{256}.$$

Consequently there is $c_H > 0$ such that, for class masses s_i sufficiently close to a, with $q = \sum_i s_i$ and T_s the corresponding Clebsch step graphon,

$$F(W_0) - F(T_s) \geq c_H \sum_i (s_i - q/16)^2. \tag{78}$$

Proof. Let A be the Clebsch adjacency matrix. The counts above give

$$P(\mathbf{1}) = 192, \qquad \nabla P(\mathbf{1}) = 60\mathbf{1}, \qquad \nabla^2 P(\mathbf{1}) = 12(J - I + A),$$

whereas $E(\mathbf{1}) = 40$, $\nabla E(\mathbf{1}) = 5\mathbf{1}$ and $\nabla^2 E = A$. Hence the logarithmic Hessian is

$$\frac{12(J - I + A)}{192} - \frac{3600J}{192^2} - \frac{5}{2}\left(\frac{A}{40} - \frac{25J}{40^2}\right) = \frac{J - 16I}{256}.$$

The gradient vanishes at $\mathbf{1}$. On the hyperplane $\sum_i h_i = 0$, Taylor's formula gives

$$\log F(\mathbf{1} + h) = \log F(\mathbf{1}) - \frac{\|h\|_2^2}{32} + O(\|h\|_2^3).$$

Apply this with $h_i = 16s_i/q - 1$ and use scale invariance. Since q stays close to one, this gives (78) with a uniform positive constant. □

A.2. **Fractional row profiles.** Let $\mathcal{M} \subseteq \{0,1\}^{16}$ consist of the zero vector and the sixteen Clebsch neighborhood vectors. Extend $c(N)$ multilinearly to $p \in [0,1]^{16}$:

$$c(p) = \sum_{\substack{U \subseteq V(\mathrm{Cl}) \\ \mathrm{Cl}[U] \cong P_4}} \prod_{\substack{i \in U \\ i \text{ an endpoint}}} p_i \prod_{\substack{j \in U \\ j \text{ internal}}} (1 - p_j).$$

This is the expectation of $c(N)$ for an independent Bernoulli subset with inclusion probabilities p_i. The neighborhood lemma implies the quantitative bound

$$12 \sum_i p_i - c(p) \geq \frac{\mathrm{dist}_1(p, \mathcal{M})}{272}. \tag{79}$$

For completeness, let ϵ be the probability that the Bernoulli subset lies outside $\mathcal{M}$. The Boolean deficit is an integer and at least one there, so the left side is at least ϵ. No two members of $\mathcal{M}$ differ in a single coordinate: the distances from zero to a neighborhood are five, and between distinct neighborhoods are six or ten. Conditioning on all other coordinates therefore gives $\min(p_i, 1 - p_i) \leq \epsilon$ for each i. Round every coordinate to its nearest Boolean value, obtaining v. If $\epsilon < 1/17$ and $v \notin \mathcal{M}$, then

$$\epsilon \geq \Pr(Y = v) \geq 1 - \sum_i \min(p_i, 1 - p_i) \geq 1 - 16\epsilon > \epsilon,$$

a contradiction. In this case $\mathrm{dist}_1(p, \mathcal{M}) \le 16\epsilon$. For $\epsilon \ge 1/17$ the trivial distance bound sixteen gives 272ϵ, proving (79). In particular, the only zero-deficit profiles are the members of $\mathcal{M}$.

For a measurable row $f : [0,1] \to [0,1]$ set $p_i(f) = a^{-1} \int_{A_i} f$. Interpret each element $r \in \mathcal{M}$ as its associated step-function row. Since r is Boolean,

$$\|f - r\|_1 = a\|p(f) - r\|_1. \tag{80}$$

The pentagon link of a new vertex with row f in the perfect background W_0 is exactly $a^4 c(p(f))$, and its degree is $a \sum_i p_i(f)$. Indeed, deleting that vertex leaves an induced P_4. An induced P_4 in an independent-class blow-up uses four distinct classes, because P_4 has no false twins. Its four row factors can therefore be integrated separately. The two orientations of each such path cancel the factor $1/2$ in the rooted pentagon link.

A.3. **Continuity and variations.** The following estimates hold also on a measurable subspace of total mass at most one. The reference values are

$$E_0 = 40a^2 = \frac{5}{32}, \qquad P_0 = 192a^5 = \frac{3}{16384}, \qquad \kappa_0 = \frac{5P_0}{2E_0} = 12a^3.$$

For graphons U, V, telescoping the ten factors in the induced pentagon integrand gives

$$|P(U) - P(V)| \le \|U - V\|_1, \qquad |E(U) - E(V)| \le \tfrac{1}{2}\|U - V\|_1. \tag{81}$$

Thus $\kappa(W) = 5P(W)/(2E(W))$ is Lipschitz in a fixed neighborhood of W_0.

We also allow a measurable density $z \in [0,1]$ on the vertex measure. Write $P(W,z)$, $E(W,z)$ for the resulting integrals, $d_z(x) = \int W(x,y)z(y)\,dy$, and $L_z(x)$ for the first variation of P at a vertex. Explicitly L_z is one half the integral of the rooted induced pentagon product over the other four vertices, with their densities included. For a differentiable density path,

$$\frac{d}{dt} F(W, z_t) = E(W, z_t)^{-5/2} \int z_t'(x)[L_{z_t}(x) - \kappa(W, z_t) d_{z_t}(x)]\,dx. \tag{82}$$

If z differs from one only on a set of measure μ, uniformly in x one has

$$|L_z(x) - L_1(x)| \le 2\mu, \qquad |d_z(x) - d_1(x)| \le \mu. \tag{83}$$

The changes in P and E are at most $\mu/2$ and μ, respectively. These estimates follow by telescoping the four or five vertex density factors. Keeping the row $f = W(x,\cdot)$ fixed, but replacing the background among the other four vertices by W_0, gives

$$|L_1(x) - a^4 c(p(f))| \le 3\|W - W_0\|_1. \tag{84}$$

There are six background factors, each with the rooted prefactor $1/2$.

We need a small-degree bound that does not assume triangle-freeness:

$$L_z(x) \le \frac{1}{4} d_z(x)^2 E(W, z). \tag{85}$$

To see this, fix the middle edge bc of the rooted pentagon $x-a-b-c-d-x$. Drop the chord condition between a, d. The two remaining neighbor choices have masses

$$r = \int W(x,u)W(u,b)(1-W(u,c))z(u)\,du,$$
$$s = \int W(x,u)W(u,c)(1-W(u,b))z(u)\,du.$$

The inequality $u(1-v)+v(1-u) \le 1$ for $u, v \in [0,1]$ gives $r+s \le d_z(x)$ and $rs \le d_z(x)^2/4$. Drop the two factors involving the chords xb, xc and integrate the middle edge, retaining its factor $1/2$. This proves (85).

For a symmetric edge variation H,

$$\begin{aligned} DP(W)[H] &= \frac{1}{2}\int q_W(x,y)H(x,y)\,dx\,dy, \\ DF(W)[H] &= \frac{1}{2E(W)^{5/2}}\int (q_W - \kappa(W))H. \end{aligned} \tag{86}$$

Here q_W is the difference of two three-variable integrals: remove one cycle-edge factor from the induced pentagon product for the positive term, and remove one chord factor for the negative term. The five cycle edges and five chords, together with the factor $1/10$, give the prefactor $1/2$.

For a Boolean step graphon T put $e_x = \int |W(x,z) - T(x,z)|\,dz$ and $e = \|W-T\|_1$. Each of the two integrands defining q has three factors incident to x, three incident to y, and three among the other vertices. Telescoping yields

$$|q_W(x,y) - q_T(x,y)| \le 6(e_x + e_y + e). \tag{87}$$

By Lemma A.3,

$$q_{W_0}(x,y) - \kappa_0 = \begin{cases} 12a^3, & \text{adjacent types,} \\ -18a^3, & \text{distinct nonadjacent types,} \\ -12a^3, & \text{equal types.} \end{cases} \tag{88}$$

The pair coefficients are multiplied by a^3, the mass of each old triple. Set $g = 12a^3 > 0$. By continuity, a Clebsch step graphon with class masses sufficiently close to a retains these signs with margin at least $3g/4$.

A.4. **Proof of the local theorem.**

Proof of Theorem A.1. Choose a fixed $\eta > 0$ small enough that the η-balls around the seventeen reference rows are disjoint and

$$\frac{\eta E_0}{2} < \frac{\kappa_0}{4}, \qquad 12\eta < \frac{g}{8}.$$

We subsequently choose δ_0 sufficiently small, with η held fixed. Write $\delta = \|W - W_0\|_1$.

Remove rows with a definite link deficit. Let B consist of points whose rows have distance at least η from every reference row. Since the original template row is one of these rows,

$$\mu_B := |B| \le \delta/\eta.$$

By (79) and (80), the perfect-background deficit on B is at least $a^4\eta/17$. Along the deletion path $z_t = 1 - t\mathbf{1}_B$, equations (81)–(84) therefore give, uniformly for almost every $x \in B$,

$$L_{z_t}(x) - \kappa(W, z_t) d_{z_t}(x) \le -\frac{a^4\eta}{17} + O(\delta + \mu_B) \le -\frac{a^4\eta}{34}.$$

The implicit constant depends only on the fixed template and the fixed η. The last inequality follows by reducing δ_0. The edge mass stays bounded away from zero. By (82), deleting B increases F by at least $c_B\mu_B$ for some $c_B > 0$.

Remove near-zero rows. Among the remaining points let Z consist of rows whose original distance from zero is less than η. Their original degrees are at most η, and

$$\mu_Z := |Z| \le \frac{\delta}{5a - \eta},$$

because each original template row has degree $5a$. Delete Z after B. During this deletion its degrees are still at most η, while $E \le 2E_0$ and $\kappa \ge \kappa_0/2$ for sufficiently small δ_0. By (85),

$$L_{z_t}(x) - \kappa(W, z_t) d_{z_t}(x) \le -\frac{\kappa_0}{4} d_{z_t}(x) \qquad (x \in Z).$$

If D_Z denotes the edge mass removed in this step, then

$$D_Z = \int_0^1 \int_Z d_{z_t}(x)\, dx\, dt.$$

Thus deleting Z increases F by at least $c_Z D_Z$ for a constant $c_Z > 0$. The deletion of a measurable set is equivalently its replacement by isolates.

Assign the remaining rows to clone types. Let $M = [0,1] \setminus (B \cup Z)$. Every point of M is within η of a unique nonzero reference row. Call its type $\sigma(x)$; let $\tau(x)$ be its original part label, and put $M_i = \{x \in M : \sigma(x) = i\}$. Distinct Clebsch rows have distance at least $6a$, so the measure ν of the misassigned set $\{\sigma \ne \tau\}$ satisfies

$$\nu \le \frac{\delta}{6a - \eta}.$$

In particular, for $s_i = |M_i|$,

$$|s_i - a| \le \mu_B + \mu_Z + \nu = O(\delta). \tag{89}$$

On M let $T(x,y) = A_{\sigma(x),\sigma(y)}$ be the ideal Clebsch step graphon. For almost every $x \in M$,

$$\int_M |W(x,y) - T(x,y)|\, dy \le \eta + \nu, \qquad e_M := \|W|_{M^2} - T\|_1 \le \delta + 2\nu. \tag{90}$$

For the first inequality, the assigned reference row differs from its reclassified version only on the misassigned set. For the second, T differs from the original template on pairs having at least one misassigned endpoint.

Repair all remaining edge errors. Consider $W_t = T + t(W|_{M^2} - T)$ for $0 \le t \le 1$, on the measure space M. The row errors and global error remain bounded by the right-hand sides of (90) throughout this path. By (89), the margin for T is at least $3g/4$. Equations (81), (87) and (90) imply

$$|(q_{W_t} - \kappa(W_t)) - (q_T - \kappa(T))| \le 12\eta + O(\delta) < g/4.$$

Hence $q_{W_t} - \kappa(W_t) \ge g/2$ where $T = 1$, and it is at most $-g/2$ where $T = 0$. The variation $H = W|_{M^2} - T$ has the opposite sign on each region. From (86),

$$\frac{d}{dt} F(W_t) \le -c_E e_M$$

for a fixed $c_E > 0$. Integrating gives $F(T) - F(W|_{M^2}) \ge c_E e_M$.

Balance the classes and estimate the distance. Put $q = |M|$ and $V = (\sum_i (s_i - q/16)^2)^{1/2}$. The Hessian estimate (78) applies by (89). Combining the four comparisons gives

$$F(W_0) - F(W) \ge c_B \mu_B + c_Z D_Z + c_E e_M + c_H V^2. \tag{91}$$

Isolating B costs at most $2\mu_B$ in L^1, and then isolating Z costs exactly $2D_Z$. Replacing the remaining graphon by T costs e_M. Within M, balance the class masses by moving a set of measure $\frac{1}{2}\sum_i |s_i - q/16|$ between classes; atomlessness permits this operation. The resulting graphon belongs to $\mathcal{E}_{\mathrm{Cl}}$, and this last change costs at most $\sum_i |s_i - q/16| \le 4V$. Thus

$$d_1(W, \mathcal{E}_{\mathrm{Cl}}) \le 2\mu_B + 2D_Z + e_M + 4V.$$

Since $\mu_B, D_Z, e_M \le 1$, its square is at most $16\mu_B + 16D_Z + 4e_M + 64V^2$. Together with (91) this proves (76), after multiplying by $5^{5/2}$ and adjusting the positive constant.

If $R_{\mathrm{ind}}(W) = R_*$, all four nonnegative terms in (91) vanish. Thus B is null, Z consists of actual isolates, the remaining graphon is exactly T, and its sixteen class masses are equal. This proves the equality assertion without assuming that the distance infimum is attained. The converse follows by homogeneity. □

Computational verification and reproducibility

The computational proof consists of explicit finite rational identities. The supplementary material provides the seven-vertex graph atlas, the rooted flag spaces, primitive pattern counts, three rational witness files, and their generators and verifiers. The main certificate contains 184 positive square terms, 37 unrestricted identity columns, and 826 positive host remainders. The multipliers of $\Phi_{\triangle}$ and J_{del} are positive, while those of J_0 and J_1 are zero. Every one of the $1,044$ host identities is checked with exact integer or rational arithmetic. A second implementation contracts integer Gram

matrices against independently recounted host tensors. The equality-span witnesses are bound to the same main certificate and checked coordinate by coordinate. Numerical optimization is used only to discover witnesses; its floating-point output is not accepted as proof.

The supplementary programs also enumerate all $65,536$ Boolean Clebsch profiles and all $6,496$ small-offset presentations under the convention in Section 14.1. An independent enumeration of labeled exceptional vertices gives the same canonical family. Formula (75) is recomputed for every presentation, and the distinct polynomial formulas are additionally checked on actual finite graphs. The branch counts in Section 14.1 have a separate direct census check. These are verifications of the finite inputs and construction formulas. The global bound, stability arguments, and passage to eventual extremality are proved in the text. In particular, the computations do not provide a numerical global threshold $k_0(j)$.

The command `python3 src/verify_project.py` reruns the complete finite verification chain. Definitions, normalization conventions, dependencies, and individual commands are specified in the accompanying `README.md` and computational reproduction guide.

AI-assisted research methodology. ChatGPT (OpenAI) assisted with proof exploration, symbolic calculations, implementation of generators and verifiers, and internal mathematical cross-checks. Its mathematical output was treated as candidate reasoning, not as a certificate of correctness. The supplied deterministic programs perform the exact finite checks described above, and the analytic arguments are given explicitly in the paper.

Declaration of generative AI and AI-assisted technologies

ChatGPT (OpenAI) was used in drafting and revising the manuscript and in organizing computational documentation. Its use in mathematical exploration and code development is described in the preceding methodological statement. The author is responsible for the mathematical claims, computations, references, and final text of this article.

References

1. József Balogh, Ping Hu, Bernard Lidický, and Florian Pfender, *Maximum density of induced 5-cycle is achieved by an iterated blow-up of 5-cycle*, European Journal of Combinatorics **52** (2016), 47–58.
2. Levente Bodnár, *FlagAlgebraToolbox: Flag algebra computations in SageMath*, 2026, arXiv:2601.06590v2.
3. Levente Bodnár, Jun Gao, Jared León, Xizhi Liu, Oleg Pikhurko, and Shumin Sun, *The inducibility of 6-vertex graphs*, 2026, arXiv:2606.00290v3.
4. Levente Bodnár and Oleg Pikhurko, *Some exact inducibility-type results for graphs via flag algebras*, Electronic Journal of Combinatorics **33** (2026), no. 2, Paper No. P2.39.
5. Ting-Wei Chao, Asaf Cohen Antonir, Anqi Li, and Hung-Hsun Hans Yu, *Edge inducibility via local directed graphs*, 2025, arXiv:2509.24064v1 [math.CO].
6. Eoin Davey, Eoin Hurley, Rémi de Joannis de Verclos, Ross J. Kang, and Jan Volec, *Local flag algebras*, 2026, arXiv:2607.12461v1 [math.CO].

7. Bernard Lidický, Connor Mattes, and Florian Pfender, *C_5 is almost a fractalizer*, Journal of Graph Theory **104** (2023), no. 1, 220–244.
8. László Lovász and Balázs Szegedy, *Limits of dense graph sequences*, Journal of Combinatorial Theory, Series B **96** (2006), no. 6, 933–957.
9. ———, *Szemerédi's lemma for the analyst*, Geometric and Functional Analysis **17** (2007), no. 1, 252–270.
10. T. S. Motzkin and E. G. Straus, *Maxima for graphs and a new proof of a theorem of Turán*, Canadian Journal of Mathematics **17** (1965), 533–540.
11. Oleg Pikhurko, Jakub Sliačan, and Konstantinos Tyros, *Strong forms of stability from flag algebra calculations*, Journal of Combinatorial Theory, Series B **135** (2019), 129–178.
12. Alexander A. Razborov, *Flag algebras*, Journal of Symbolic Logic **72** (2007), no. 4, 1239–1282.
13. Yichen Wang, Xiamiao Zhao, and Mei Lu, *Edge version of the inducibility via the entropy method*, Electronic Journal of Combinatorics **33** (2026), no. 2, Paper No. P2.17, 16 pp.